

Consumption in Asset Returns*

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Abstract

Using information in returns we identify the stochastic process of consumption. We find that aggregate consumption reacts over multiple quarters to innovations spanned by financial markets. This persistent component accounts for over a quarter of consumption variation. These shocks command a large and significant risk premium, driving a large share of stocks and a small yet significant fraction of bonds time series variation. Nevertheless, we find no support for stochastic volatility of consumption driving time-varying risk premia. Finally, an otherwise standard recursive utility model based on our estimated process explains equity premium and risk-free rate puzzles with low risk aversion.

Keywords: Consumption Dynamics, Asset Returns, Consumption-Based Asset Pricing, Term Structure.

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Consumption-based models have contributed to our understanding of financial markets, business cycle dynamics, and household decision-making. In these settings, assumptions about consumption persistence and volatility have a profound impact on both empirical performance and policy implications of models. However, using consumption data alone, it is difficult to identify the underlying stochastic process.¹ As a result, macro-finance researchers tend to rely on assumptions that are difficult, if not impossible, to validate outside the frameworks under consideration. Chen, Dou, and Kogan (2021) refer to this source of model fragility as “dark matter.”

As we show, if the stochastic process of consumption were of the type usually postulated in the literature, standard methods would fail to accurately recover the magnitude and persistence of its conditional mean and volatility – crucial ingredients for asset pricing. Instead, we develop a novel state-space approach that sheds light on the dark matter of consumption-based models. We find that consumption growth is highly predictable at business cycle frequency by shocks spanned by financial markets that account for *more than a quarter* of its variance. Furthermore, these shocks to the conditional consumption mean command a large and significant risk premium in the cross-section of asset returns. Nevertheless, we find no support for the stochastic volatilities of consumption driving time-varying risk premia. Our findings hold across a range of reasonable measures of consumption growth, choices of base assets, specifications of external predictors, assumptions about mismeasurement in consumption, and misspecification of the model.

Our identification strategy is rooted in the central insight of the intertemporal Euler equation of models that have consumption as one of the state variables in the utility function: Most shocks affecting the household force it to adjust *both* investment and consumption plans. This is exactly the insight that motivated the modeling of the conditional moments of consumption as a function of information in stocks and bonds, for example, in Harvey (1988) and Kandel and Stambaugh (1990). In fact, asset prices that reveal information about the state variables of the economy are a feature of almost any consumption-based macro-finance model. Therefore, we use the cross-section of returns to extract innovations reflected in both consumption and financial assets. In addition, our identification strategy is confirmed

¹See, e.g., Beeler and Campbell (2012), Campbell (2017), Cochrane (2005), and Ludvigson (2013) for a review of the empirical challenges of consumption-based asset pricing.

by data on shareholders' and non-shareholders' consumption: Only the consumption of the former reacts to financial market shocks, with the aggregate data being in between the two.

Our approach allows the joint consumption and return data to speak for itself and establishes a new set of facts regarding the dynamics of stocks, bonds, and consumption growth. Conceptually, the method is simple. As in Blanchard and Quah (1989), we leverage the fact that any covariance stationary process for consumption, including but not limited to those used in habits and long-run risk models, can be represented by a Moving Average (MA). As in the structural vector autoregression literature (e.g., Christiano, Eichenbaum, and Evans (2005) and Sims and Zha (2006)), we impose an economic restriction to identify the impulse response of consumption to shocks spanned by financial markets, and, hence, reliably estimate its conditional mean and persistence. This, in turn, allows us to correctly recover the volatility dynamics, because the latter, as we show, can be consistently estimated *only if* the conditional consumption mean and its predictability are properly captured. Furthermore, the MA modeling is also motivated by measurement problems in consumption data, such as time aggregation (Breedon, Gibbons, and Litzenberger (1989)) and benchmarking (Wilcox (1992)), that we formally account for in our analysis.

The crucial role of the conditional mean in driving consumption growth that we uncover (over 25% of total consumption variance) is more than twice what is normally assumed in leading macro-finance models. For instance, the conditional mean contribution to consumption variance is zero in Lucas (1987) that calculates the cost of business cycle fluctuation as well as in models that, for parsimony, do not encode any predictability in consumption (e.g., the habit framework of Campbell and Cochrane (1999) and the rare disasters model of Barro (2006)). It is instead about 12% in the long-run risk calibrations of Bansal and Yaron (2004) and Hansen, Heaton, Lee, and Roussanov (2007) (to the best of our knowledge, the largest used in the literature).

The impact of shocks to the consumption growth mean is well identified, sharply estimated, and economically large: A one-standard-deviation innovation implies a response of consumption growth of about 1% over the next two years. Moreover, it takes about two years for the conditional mean shocks to be fully reflected in the consumption process, producing substantial predictability, and no further reaction is apparent after this horizon.

The consumption shocks spanned by financial markets generate a clear business cycle pattern in the conditional expectation of consumption growth, have a large and significant risk premium in the cross-section of asset return, with an implied annualized Sharpe ratio of about 0.5. This in turn implies, as we show, that an otherwise standard Epstein and Zin (1989) preference setting calibrated with our estimated process can match the market equity risk premium with a low level of risk aversion: Point estimates range from 6.5 to 27, with values as low as 10 always within the 90% confidence intervals. Furthermore, assuming an elasticity of intertemporal substitution above one, the model can also match the level and volatility of the risk-free rate. Nevertheless, we uncover a challenge for macro-finance modeling. Often, representative agent models obtain equilibrium time-varying risk premia by postulating stochastic volatilities (SVs) in consumption, which leads to a dependence of excess returns on the conditional volatilities. We find no evidence supporting this mechanism, even after controlling for time-averaging bias in the consumption process. However, we detect a significantly positive comovement between the squared VXO index and the stochastic variance of the consumption mean shock, with a posterior mean correlation coefficient of about 45%.²

Crucially, all of our findings are obtained without restricting either the nature or number of factors driving asset returns, allowing for general and flexible SV processes in both consumption and returns (with short- and long-run SVs, as in Schorfheide, Song, and Yaron (2018), as well as the leverage effect), and hold using quarterly, monthly, and mixed-frequency estimations.

Estimation consistency of a volatility process crucially depends on the correct specification of the conditional mean, especially when the latter is persistent and difficult to detect, as postulated in leading models. For this reason, the key driver of our findings is the correct identification of the conditional mean process of consumption. Following the insight of Jagannathan and Marakani (2016), we show that using consumption data alone is not enough to reliably identify the consumption process, and we leverage instead the information in a rich cross-section of asset returns. Similar to Schorfheide, Song, and Yaron (2018),

²The VXO is an implied volatility index for the S&P 100 single-name stocks produced by the Chicago Board Options Exchange. We employ VXO, rather than the more canonical VIX, due to its longer data series and very high correlation with the latter.

we use a state-space approach to filter the latent shocks to the consumption process. But unlike them, we do not constrain the consumption mean process to a stringent parametric functional form.

Our econometric framework is flexible and comprehensive. We model consumption growth as the sum of two independent processes: a high-order MA that (potentially) co-moves with returns and a transitory component orthogonal to financial assets (and external predictors). Innovations to returns are, in turn, modeled as depending (potentially) on the shocks to the persistent component of consumption, potentially external predictors, and a set of orthogonal latent factors that drive the residual comovement among assets. As in Wold’s decomposition, we estimate the MA representation of consumption (equivalently, its impulse response). Nevertheless, unlike in the existing literature, we identify the MA innovations by using a large cross-section of asset returns. These economic innovations are allowed to follow distinct SV processes, which can further act as predictors for asset returns, as in many equilibrium macro-finance models.³

With extensive simulations, we show that our method, even in samples as small as the historical sample, robustly recovers the consumption dynamics. This is a difficult task, and indeed we find that popular model selection procedures (e.g., AIC and BIC criteria) fail to identify the type of consumption processes postulated in the literature. Rather, our approach correctly recovers the conditional mean of consumption growth and, therefore, its volatility process. As we show, missed and/or misspecified predictability in the conditional mean process generates spurious volatility dynamics similar to what is documented in the existing literature.

³Our general MA formulation for the consumption process allows shocks to be potentially driven by a jump process. However, in the empirical estimation we model all innovations via continuous distributions. That is, we do not explicitly model the large jumps often used to calibrate rare disaster models. Nevertheless, note that these typical calibrations (e.g., Barro (2005), Wachter (2013)) are based on the assumption that the cumulative disasters, recorded in the data over multiple years, take place during a single year (for a detailed discussion of this counterfactual assumption, see Constantinides (2008) and Julliard and Ghosh (2012)). At the quarterly and monthly frequencies we focus on, properly taking into account the time series properties of disasters would imply modelling quite small jumps. For instance, the largest recorded contraction in quarterly nondurable consumption per capita during the Great Recession, -1.7% in 2008:Q4 (included in our sample), is less than 2.4 standard deviations of consumption growth. Similarly, even the cumulative contraction in U.S. consumption during the Great Depression (17%, that is, about half of the cumulative GDP drop of 31% often used in calibrations) implies average quarterly consumption contractions of about 1% per period. Given the frequency of our data, these small jumps can be well approximated by, and are hard to distinguish from, a continuous process (see, e.g., Aït-Sahalia (2004)).

The key identifying restriction we rely on is that innovations in consumption spanned by financial markets should be reflected in returns at the same time as they occur. Instead, consumption could adjust slowly to these shocks. This assumption is not only theoretically sound (see, e.g., Caballero and Simsek (2023)), since equilibrium prices are jump variables, but is also supported by a rich set of reduced-form empirical evidence in the previous literature (that we elaborate on in the Internet Appendix D). In addition, our formulation implies a particular term structure for the covariances between asset returns and multi-period consumption growth. We measure this term structure directly in the data and find that it yields the same size and persistence of the conditional mean process as in our state-space setting. Moreover, our estimated conditional consumption mean is a predictor of future consumption growth superior to other commonly used proxies (e.g., lagged consumption growth, its AR or VAR forecasts, lagged GDP growth, the Survey of Professional Forecaster, and the Liu and Matthies (2022) news index).

The central feature of our framework is that we do not impose stringent parametric restrictions on the functional form of the conditional consumption mean process. Our approach is by construction robust to the nature and number of factors driving asset returns, the priced or unpriced nature of the shocks, and the role of external predictors. It allows us to compare and test not only specific structural models but also entire classes of economic mechanisms. To the best of our knowledge, this yields the most comprehensive analysis of the stochastic process for consumption to date and should guide both the design and calibration of macro-finance models.

Closely Related Literature

By its very nature, our paper is closely linked to consumption-based asset pricing.⁴ Given both the saliency of consumption dynamics for equilibrium macro-finance models and the ambiguous nature of the true data-generating process, akin to the dark matter in finance (Campbell (2017) and Chen, Dou, and Kogan (2021)), much effort has been put into deter-

⁴For example, Breeden, Gibbons, and Litzenberger (1989), Lettau and Ludvigson (2001, 2001b), Jacobs and Wang (2004), Hansen, Heaton, Lee, and Roussanov (2007), Jagannathan and Wang (2007), Piazzesi, Schneider, and Tuzel (2007), Constantinides and Ghosh (2017), and Bansal, Kiku, and Yaron (2012). See also Ludvigson (2013) for an excellent review.

mining its key features. Hence, our paper is naturally related to several large strands of the literature.

Proper identification and recovery of the consumption process, as well as its link to asset returns, is known to be an elusive problem. Consumption-based asset pricing models are notoriously difficult to estimate due to the weak identification of the underlying parameters. This problem is particularly pronounced in the context of linear factor models (Burnside (2011, 2015), Kan and Zhang (1999), and Kleibergen and Zhan (2020)) that consider consumption growth as a source of priced risk in the cross-section of asset returns. The challenge of identifying the stochastic process of consumption growth has also been highlighted in the structural models of consumption-based asset pricing. In particular, Johannes, Lochstoer, and Mou (2016) and Collin-Dufresne, Johannes, and Lochstoer (2016) illustrate the problem from the perspective of a Bayesian learner and show that it has a profound impact on equilibrium consumption and return dynamics.

Our solution to the identification problem of consumption dynamics relies on insights from two strands of literature in empirical macroeconomics. First, following Blanchard and Quah (1989), we rely on the MA representation of the stochastic process. Second, as in the structural VAR literature (e.g., Christiano, Eichenbaum, and Evans (2005) and Sims and Zha (2006)), we use the intertemporal Euler equation to formulate a constraint on the joint dynamics of asset returns and consumption and identify their common shocks and propagation mechanisms. This allows us to trace out the response of consumption to shocks spanned by financial market as in Parker (2001), but without assuming that return innovations are fully captured by the market index and that consumption reacts only with a lag to financial shocks.

Using data from the Panel Study of Income Dynamics (PSID), we show that the dynamics uncovered in aggregate data also hold for shareholders but – importantly – not for non-shareholders, further supporting our identification strategy and rationalizing the findings of Malloy, Moskowitz, and Vissing-Jorgensen (2009). Finally, our approach is also supported by the evidence in Ang, Piazzesi, and Min (2006), who show that financial markets are informative about macro dynamics, and Liew and Vassalou (2000), who find that the equity value and size factors are leading indicators of economic growth.

We follow the logic of Jagannathan and Marakani (2016) and leverage a large cross-section of equity and bond returns to amplify the signal common to consumption and asset returns. The high-dimensional nature of the problem, therefore, naturally calls for the use of Bayesian filtering techniques. Similar to Schorfheide, Song, and Yaron (2018) and Zviadadze (2021), we leverage state-space modeling to infer the latent dynamics of consumption and returns. Unlike them, however, we do not rely on the restrictive assumption of a low-order autoregressive structure (e.g., AR(1)), because this would constrain the *entire* long-run dynamics of the conditional mean of consumption (Beeler and Campbell (2012)). This element is key, because misspecifying the conditional mean gives rise, as we show, to spurious volatility dynamics in consumption.

Our study is also related to a recent strand of papers that employ Bayesian tools to evaluate the assumptions, long considered to be salient, yet largely untestable by the frequentist literature. Giannone, Lenza, and Primiceri (2021) rely on the spike-and-slab priors to test the cornerstone assumption of sparsity in a variety of applications. Bryzgalova, Huang, and Julliard (2023) leverage the hierarchical structure implied by the no-arbitrage constraint in cross-sectional asset pricing to parse the factor zoo and test the underlying assumptions of uniqueness and factor sparsity of linear models. Our flexible modeling allows us to compare entire classes of data-generating processes for consumption and returns, postulated (and not) in the prior literature. As a result, we are able to test the key assumptions – usually taken for granted – commonly used to generate time-varying risk premia in macro-finance models.

The low power of frequentist approaches in consumption-based models often forces researchers to rely on proxies of consumption volatility. Bansal, Kiku, Shaliastovich, and Yaron (2014) and Campbell, Giglio, Polk, and Turley (2018) use realized volatility of returns as a proxy for that of consumption growth, motivated by the market clearing condition of a closed Lucas tree economy with no labor or public spending. Dew-Becker and Giglio (2016) further acknowledge the lack of sufficient power in testing whether stochastic volatility drives time-varying risk premia. Our framework, instead, allows us to estimate the volatilities of consumption and assets and test their persistence, equality, and other features. We find that the data reject the notion of equality between consumption and return volatilities and indicate that neither of them drives time variation in conditional risk premia.

Our work is also connected to a large body of literature on robust inference in potentially misspecified models (Hansen and Sargent (2001)). Similar to Giglio and Xiu (2021), we allow for multiple latent and observable factors to drive a cross-section of asset returns. Our specification for the consumption dynamics is also agnostic about the exact functional form consistent with the underlying MA decomposition, treating the consumption data-generating process as unknown. In the consumption habit setting, a similar approach has been undertaken by Chen and Ludvigson (2009), who, rather than postulating a particular functional form of habit, provide its general nonparametric estimate.

Finally, our paper also contributes to the literature that studies the link between low-frequency consumption movements and financial returns (Lettau and Ludvigson (2001, 2001b), Parker and Julliard (2005), Jagannathan and Wang (2007), Ortu, Tamoni, and Tebaldi (2013), Dew-Becker and Giglio (2016), and Bandi and Tamoni (2023)). We identify the stochastic process of consumption by (primarily) tracing out the term structure of its response to shocks spanned by financial markets. In doing so, we document that it is the same consumption shocks, weakly identified in the short run (as in Kleibergen and Zhan (2020)), but economically and statistically salient at business cycle frequency. Furthermore, we show that this mechanism is unlikely to be driven by measurement error, time-averaging and benchmarking of consumption data (Hansen and Singleton (1983), Breeden, Gibbons, and Litzenberger (1989), and Wilcox (1992)), as our results also hold with mixed-frequency estimation, monthly data, formally accounting for observational error, and in a broad spectrum of robustness exercises.

Although our method cannot distinguish the economic mechanisms driving the persistency of the conditional consumption mean (e.g., adjustment costs in consumption and complementary factors in the marginal utility, or constraint on information processing), we do pin down the process for consumption (and asset returns) that a valid macro-finance model should either deliver in equilibrium or that a researcher should at least use for a realistic calibration of the data-generating process.

The remainder of the paper is organized as follows. Section I demonstrates that the types of consumption process that are often used in the macro-finance literature are unlikely to be correctly identified using canonical model specification selection. Our state-space

formulation is introduced in Section II, where we also show that it can accurately recover the consumption dynamics of popular models, as well as more general formulations. Our main empirical findings are presented in Section III, Section IV outlines their main implications for equilibrium asset pricing models, and Section V concludes.

I The Challenge of Consumption Persistency

We begin by showing that the type of consumption processes assumed in most macro-finance models is difficult to detect and, hence, test in samples of the same size as historical samples.

Consider, for example, the so-called long-run risk process of Bansal and Yaron (2004). In this formulation, log-consumption growth, $\Delta c_{t,t+1}$, contains a persistent AR(1) component, x_t , which is crucial in rationalizing unconditional risk premia and other moments of the historical data. That is,

$$\Delta c_{t,t+1} = \mu + x_t + \sigma_t \eta_{t+1}, \quad x_{t+1} = \rho x_t + \phi_e \sigma_t e_{t+1}, \quad (1)$$

where $\eta_t, e_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and σ_t , depending on the calibration, is either a constant or a SV process. The amount of time-series variation of consumption that is driven by the predictable component x_t varies in the literature. For instance, the conditional mean generates about 12% of the quarterly consumption variance in Bansal and Yaron (2004) and Hansen, Heaton, Lee, and Roussanov (2007).

But would a researcher be likely to detect this persistency and identify its functional form in samples of the same size as the historical sample? We address this question formally by using the Hansen, Heaton, Lee, and Roussanov (2007) calibration to generate a monthly series of consumption of the same length as the postwar sample we use in our empirical analysis (214 quarters) and formally perform ARIMA model selection for the consumption process. We consider consumption aggregated to the quarterly frequency – as in real-world data – as well as the monthly observations that are generally much noisier in reality. We report the details of the simulation design in Internet Appendix A.

Table 1 presents the frequency of specification selected according to the Bayesian and Akaike Information Criteria (BIC and AIC). Strikingly, with both quarterly and monthly sampling, the most often selected specification implies no predictability at all (Columns

(A) and (B)). Moreover, note that this result arises using the calibration of the long-run risk process that has the *largest* predictability of consumption. Furthermore, the table shows that even if a researcher were to observe the conditional mean of consumption growth directly (Columns (C) and (D)), canonical specification selection would fail to identify the true mean process with more than 92% probability.⁵

Table 1: ARIMA model selection of the long-run risk consumption process

(A) BIC: $\Delta c_{t,t+1}$				(B) AIC: $\Delta c_{t,t+1}$				(C) BIC: x_{t+1}				(D) AIC: x_{t+1}			
Panel A: Quarterly frequency (214 observations)															
p	d	q	freq	p	d	q	freq	p	d	q	freq	p	d	q	freq
0	0	0	53.7%	0	0	0	25.1%	0	1	0	50.0%	0	1	1	21.5%
0	1	1	24.8%	0	1	1	14.6%	0	1	1	15.6%	0	1	0	13.7%
1	0	1	5.9%	1	0	1	13.8%	1	0	1	12.2%	1	0	1	11.6%
1	1	1	5.1%	0	0	1	6.3%	2	0	0	10.1%	2	0	0	7.6%
1	0	0	3.9 %	1	1	1	5.3%	1	1	0	5.1%	1	1	0	7.5%
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
Panel B: Monthly frequency (642 observations)															
0	0	0	55.1%	0	0	0	27.8%	0	1	0	92.3%	0	1	0	70.0%
0	1	1	31.3%	0	1	1	24.0%	1	0	0	7.2%	2	1	2	4.9%
1	0	1	4.7%	1	0	1	13.4%	1	1	0	0.3%	1	0	0	4.7%
5	1	0	1.4%	0	0	1	6.5%	1	1	1	0.1%	1	1	0	3.8%
0	0	1	0.9%	0	1	2	3.2%	2	0	0	0.1%	1	1	1	3.6%
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Empirical frequencies of ARIMA(p,d,q) models selected by Bayesian information criterion (BIC) and Akaike information criterion (AIC) in 1,000 simulations of the Hansen, Heaton, Lee, and Roussanov (2007) long-run risk specification for the consumption growth process. We list only the top five most frequent models.

Note that misidentifying the conditional mean process also has important consequences for recovering the volatility process. For instance, suppose that the process in equation (1) has a constant volatility; that is, $\sigma_t = \sigma \ \forall t$. In this case, if a researcher were to conclude that there is no predictability in consumption growth (as Table 1 suggests as the most likely outcome), she would then find evidence of time-varying volatility in consumption, because its squared forecast errors would be positively autocorrelated (with the j -th autocorrelation proportional to ρ^{2j}). More generally, missing the true degree of predictability in the conditional mean process mechanically delivers spurious (if the true process is homoskedastic) or biased (if the true process has time-varying volatility) evidence of volatility clustering.

⁵ Similarly, Colacito and Croce (2011) show, with a Monte Carlo simulation exercise, that recovering the consumption dynamic of equation (1) using a cross-section of countries is a daunting task when using only consumption data, even in a long time series sample.

If the true process for consumption is similar to what we rely on in macro-finance models, the considerations above imply three crucial requirements for reliable inference on the data-generating process of consumption. First, we need an estimator of the conditional mean of consumption that can capture the true degree of predictability *without* relying on fragile specification selection. This is crucial, in particular, if one wants to make statements about stochastic volatility in consumption (its existence, magnitude, and properties). Second, we should not achieve identification via arbitrary (and often non-testable within a model) parametric restrictions for the consumption process, for example, an arbitrary AR(1) process for its persistent component or the often employed proportionality restriction between consumption and return stochastic volatilities. Given their crucial role in model predictions, these restrictions should be tested whenever possible. Third, ideally, we would use a method that allows us to learn about the consumption process by leveraging information in other variables that should be adapted to the same type of shocks that drive consumption (e.g., wealth shocks). These three properties above are exactly what our empirical formulation delivers. We present it in the next section.

II A Model of Consumption and Returns Dynamics

In macro-finance models, the stochastic discount factor is typically a function of consumption growth and potentially additional variables (e.g., habits, returns on wealth, leisure, leverage ratios, and aggregation weights in heterogeneous agent models). Furthermore, consumption and returns both contribute to the intertemporal budget constraint. This implies that, with the exclusion of exceptional examples, there is a set of shocks with respect to which both returns and consumption growth are adapted processes.

The reason for this general feature of equilibrium models is that households react to shocks (e.g., wealth, income, and beliefs) by adjusting *both* consumption and investment decisions. Hence, in principle, one could leverage the information in equilibrium asset returns to learn about the shocks that drive consumption and the form of its stochastic process. This simple insight lies at the core of our empirical strategy.

In particular, to model parametrically the reaction of consumption to the same shocks

that are spanned by asset returns, we postulate that the consumption growth process can be decomposed in two terms: A serially uncorrelated disturbance, w_c with variance σ_c , which is independent of financial market shocks, and a possibly autocorrelated process – a persistent component – that possibly depends on the current and past shocks to asset returns. For expositional simplicity, we start by focusing on a setting with constant volatilities and generalize our framework to incorporate stochastic volatilities in all the shocks.

To avoid taking an ex-ante stand on the particular time-series structure of the persistent component (or its absence), we work with its (possibly infinite) MA representation. Obviously, by virtue of Wold’s representation theorem, an $MA(\infty)$ model for the persistent component would capture the true data-generating process and avoid the fallacy of model selection outlined in the previous section. Because the MA coefficients in the Wold representation are square summable, any finite order covariance stationary ARIMA can be approximated using a high-order MA process, with the accuracy increasing with the MA order. Therefore, we model the (log) consumption growth process as

$$\Delta c_{t-1,t} = \mu_c + \underbrace{\sum_{j=0}^{\bar{S}} \rho_j f_{t-j}}_{MA(\bar{S})} + w_t^c, \quad (2)$$

where $\bar{S}$ is a large positive integer (potentially equal to $+\infty$), μ_c is the unconditional mean, the ρ_j coefficients are square-summable, and, most importantly, f_t (a white noise process normalized to have unit variance) is the fundamental innovation upon which all asset returns load contemporaneously.

This representation encompasses most leading consumption-based asset pricing models. In particular, setting $\rho_j = w_t^c = 0 \ \forall j > 0$ and letting f_t to be an i.i.d. Gaussian process yields the formulation of Campbell and Cochrane (1999). The two-state Markov chain process of Mehra and Prescott (1985) is equivalent to an AR(1) process, and, hence, $MA(\bar{S} = \infty)$ and $w_t^c = 0$. In the rare disaster formulation of Barro (2005), $\rho_j = 0 \ \forall j > 0$, f_t is an i.i.d. jump variable, and w_t^c are i.i.d. shocks. Wachter (2013) shares the same formulation but with varying jump intensity. The AR(1) process in Vasicek (1977) maps into $MA(\bar{S} = \infty)$ and $w_t^c = 0$. Finally, adding an uncorrelated w_t^c shock to the Vasicek (1977) formulation results in the consumption process of Campbell (1999) and Bansal and Yaron (2004). Nevertheless, in

our empirical estimation, we restrict all innovations to follow Gaussian processes. Hence, we do not formally model large and infrequent jumps as considered in the rare disaster literature (e.g., Wachter (2013) and Julliard and Ghosh (2012)).

Consistent with these models, we assume that returns load on f_t contemporaneously:

$$\mathbf{r}_t^e = \boldsymbol{\mu}_r + \boldsymbol{\rho}_f^r f_t + \boldsymbol{\rho}_g^r \mathbf{g}_t + \mathbf{w}_t^r, \quad f_t \perp \mathbf{g}_t, \quad \mathbf{g}_t \sim \mathcal{N}(\mathbf{0}_{K-1}, \mathbf{I}_{K-1}), \quad (3)$$

where $\mathbf{r}^e$ denotes a vector of log excess returns, $\boldsymbol{\mu}_r$ is a vector of expected values, $\boldsymbol{\rho}^r$ contains the asset-specific loadings on the common risk factor, $\mathbf{g}_t$ is the additional $K - 1$ latent factors driving only asset returns but not appearing in the consumption growth equation, and $\mathbf{w}_t^r$ is uncorrelated with all other shocks and has a diagonal covariance matrix $\boldsymbol{\Sigma}_r$.⁶ The key identification assumption in equation (3) is that f_t is the only common component between consumption growth and asset returns and is orthogonal to the remaining factors $\mathbf{g}_t$. In the formulation above, returns are modeled as reacting contemporaneously and fully to the f shocks since prices are “jump” variables in equilibrium models. Note that, as shown below, equations (2)–(3) can also accommodate external predictor variables and, as we demonstrate later on, a direct dependency on the stochastic volatility processes of *all* shocks.

The joint dynamics of consumption and returns postulated in equations (2)–(3) are consistent with the extensive preliminary evidence that we report in Internet Appendix D. Therein, based on predictive regressions, we show that: *i*) the consumption growth process shows significant predictability over multiple years and *ii*) this predictability is better captured by lagged returns than by lagged consumption.

Obviously, we cannot feasibly use an infinite number of lags in the MA component of consumption growth. At the same time, because the MA representation of the persistent component is meant to approximate only the true latent dynamic, model selection would not be appropriate and, as outlined in Section I, would possibly be unreliable. Note also that employing an excessively high number of lags in the MA does *not* affect the consistency of the estimation, but only its efficiency. Consequently, in our empirical analysis, we rely on a conservative approach and: *i*) in the baseline estimation, we use up to 3.5 years of lagged

⁶The diagonality and time-invariance assumption can be relaxed, as explained in Internet Appendix B.2, and we do so in empirical applications. We will later allow this covariance to embed both common and idiosyncratic stochastic volatilities. Note also that the violation of the diagonality restriction affects only efficiency, not consistency, of the estimation of the conditional mean processes for consumption and returns.

quarterly shocks in the conditional mean of consumption because this is almost twice as much the degree of persistency that we find in the preliminary evidence in Internet Appendix D, and it covers the span of predictability uncovered in the previous literature (see, e.g., Liew and Vassalou (2000), Parker and Julliard (2005), and Bandi and Tamoni (2023)); *ii*) we show in the simulation in Section II.1 that the approach is robust to the precise number of included MA lags; and *iii*) we confirm that all of our empirical results are virtually identical even when including up to 12.5 years of past quarterly shocks (i.e., 50 lags).

The dynamic system in equations (2)–(3) can be reformulated as a state-space model, and Bayesian inference can be conducted to estimate both the unknown parameters $(\mu_c, \boldsymbol{\mu}_r, \{\rho_j\}_{j=0}^{\bar{S}}, \boldsymbol{\rho}^r, \sigma_c^2, \Sigma_r)$ and the time-series of the unobservable common factor of consumption and asset returns $(\{f_t\}_{t=1}^T)$. This estimation procedure is described in Internet Appendix B.

A crucial point that allows us to identify the shocks is the lead-lag structure of the consumption process and its possible link to asset returns. Without equation (2), the shocks would be under-identified, making it difficult to give any particular rotation a structural interpretation. Another interpretation of this estimation approach is that of uncovering the shocks that drive financial returns through the impulse response function on consumption, in the spirit of the Uhlig (2005) identification in structural-VARs. In particular, our approach is akin to constructing the Generalized Impulse Response Function of consumption and financial markets, building upon the insights of Koop, Pesaran, and Potter (1996) and Pesaran and Shin (1998).

In fact, it is easy to see that the ρ_j coefficients identify the impulse response function (IRF) of multiperiod consumption growth to the shock f_t as⁷

$$\frac{\partial \mathbb{E}[\Delta c_{t-1,t+S}]}{\partial f_t} = \sum_{j=0}^S \rho_j, \quad \rho_{j>\bar{S}} := 0. \quad (4)$$

The framework described in equations (2)–(3) is closely related to Giglio and Xiu (2021), who propose a three-pass estimator of factors’ risk premia. Specifically, their paper projects the nontradable factor onto the space of asset returns’ latent factors. Their building block is that the projection of any variable onto the asset space is rotation invariant; hence, they can

⁷This immediately follows from the observation that, since $\Delta c_{t-1,t+S} \equiv \sum_{j=0}^S \Delta c_{t-1+j,t+j} \equiv \ln(C_{t+S}/C_{t-1})$, we have $[\Delta c_{t-1,t}, \Delta c_{t-1,t+1}, \dots, \Delta c_{t-1,t+S}]' \equiv \Gamma [\Delta c_{t-1,t}, \Delta c_{t,t+1}, \dots, \Delta c_{t-1+S,t+S}]'$, where Γ is a lower triangular square matrix of ones (of dimension S).

recover latent factors of asset returns via Principal Component Analysis (PCA) and regress the nontradable factor on the first several principal components (PCs). Similar to their paper, we extract the common component in nontradable factors and asset returns relying on the rotation invariance of latent factors. We show in Section II.2 that if the true f_t is a linear combination of large principal components of $\mathbf{r}_t^e$, our framework can identify it.

Furthermore, our paper nests in and improves the framework in Giglio and Xiu (2021). We allow the lead-lag relationship between nontradable factors and asset returns, whereas Giglio and Xiu (2021) study only the contemporaneous covariance structure between these variables. Because most nontradable factors, especially macro variables, are persistent to a certain extent, the flexibility of our approach can improve the identification of the spanned component between nontradable economic fundamentals and asset returns.

Finally, note that the formulation in equations (2)–(3) can be generalized to *i*) allow for a bond-specific latent factor (b_t) to which consumption could react slowly over time and *ii*) external predictor variables, $\mathbf{x}$, predetermined at time t , as follows:

$$\Delta c_{t-1,t} = \mu_c + \sum_{j=0}^{\bar{S}} \rho_j f_{t-j} + \sum_{j=0}^{\bar{S}} \theta_j b_{t-j} + \gamma_x^c \mathbf{x}_{t-1} + w_t^c \quad \text{and} \quad (5)$$

$$\mathbf{r}_t^e = \underbrace{\boldsymbol{\mu}_r}_{N \times 1} + \underbrace{\boldsymbol{\rho}^r}_{N \times 1} f_t + \left[\underbrace{(\boldsymbol{\theta}^b)'}_{N_b \times 1}, \underbrace{\mathbf{0}'_{N-N_b}}_{(N-N_b) \times 1} \right]' b_t + \underbrace{\boldsymbol{\rho}_g^r}_{N \times 1} \mathbf{g}_t + \gamma_x^r \mathbf{x}_{t-1} + \underbrace{\mathbf{w}_t^r}_{N \times 1}, \quad (6)$$

where N_b is the number of bonds that are ordered first in the vector $\mathbf{r}_t^e$, and $\boldsymbol{\theta}^b \in \mathbb{R}^{N_b}$ contains the bond loadings on the factor b_t – a white noise process with variance normalized to one.

Two observations regarding our parametric framework warrant mentioning. First, both the one-factor (equations (2)–(3)) and two-factor (equations (5)–(6)) models are overidentified. Second, the estimation of the model assuming constant volatility is generally consistent even in the presence of time-varying volatility in the true processes; hence, our formulation is robust along this dimension. We address this issue formally in Section III.2, where we show that misspecification of the consumption mean process leads to spurious evidence of consumption volatility clustering, and in Section III.6, where we generalize our state-space model to allow for stochastic volatilities affecting all the shocks in the system.

II.1 The challenge of consumption persistency redux

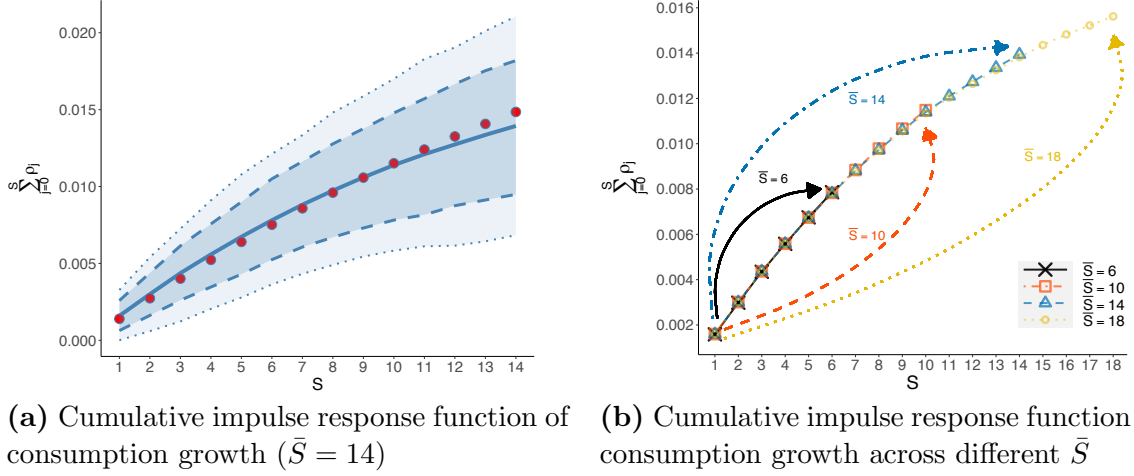
A natural question is whether our state-space representation of consumption and returns in equations (2)–(3) is able to recover the consumption process when standard methods fail (as shown in Section I). To do so, we again use the simulated Hansen, Heaton, Lee, and Roussanov (2007) long-run risk consumption process described in Section I, calibrate the asset loadings on f to the values observed in the historical data, and apply our state-space estimation to it. Note that this simulation is particularly challenging for two reasons. First, the simulated time-series sample is small, with only 200 quarterly observations. Second, consumption and return data are generated at a monthly frequency and then aggregated to the quarterly frequency. Hence, the simulated data are affected by time-averaging, making the true conditional mean much harder to recover, as illustrated in Schorfheide, Song, and Yaron (2018).

As shown in Panel (a) of Figure 1, our state-space model with a long consumption MA accurately captures the effect of time- t shocks on subsequent consumption growth: The difference between the mean estimate (across simulated time-series) and the true value is extremely small, as is the variability across simulated samples. If anything, we observe a small attenuation bias in the long run, implying that our approach is, in fact, conservative in estimating the true extent of consumption predictability. That is, the conditional mean of consumption is accurately captured by the state-space representation method. In addition, Figure IA.19 in Internet Appendix P demonstrates that our estimation precisely recovers the loading of asset returns on the shocks to the conditional mean of consumption.

Furthermore, as shown in Panel (b) of Figure 1, the IRF estimates are almost identical when using different orders for the MA component. The only difference is that with a longer MA, we can trace the effect further in the future. Hence, a finite order MA yields a conservative measure of the long-run effect of time- t shocks.

Recall also that there is a one-to-one mapping between IRFs (or, equivalently, MA representation) and variance decomposition. Hence, our accurate IRF estimates imply that we can perform accurate variance decomposition for both consumption and returns, as we do in later sections.

Figure 1: Cumulative impulse response function of consumption growth to one-standard-deviation shock to the conditional mean of consumption growth in 1,000 simulations.



Panel (a) plots the mean, 5th, 16th, 84th, and 95th percentiles of cumulative IRF in 1,000 simulations. The model is estimated under the assumption that $\bar{S} = 14$. Red dots denote pseudo true values, blue solid line denotes the mean estimates across simulations, while shaded areas denote 68% and 90% confidence intervals. The simulated time-series sample size is 200 (quarters). Panel (b) plots the average cumulative IRF of consumption growth to one-standard-deviation shock to the conditional mean of consumption growth in 1,000 simulations, using MA representation with different maximum number of lags: $\bar{S} = 6, 10, 14$, and 18.

II.2 Identifying the conditional dynamic of $\Delta c_{t-1,t}$

Section II.1 confirms that our framework can accurately recover the conditional dynamics of the consumption process in a single-factor model, where the shock to the consumption conditional mean is the only common driver in consumption and asset returns. However, some may be concerned that our approach mechanically identifies f_t as the largest principal component of returns. To address this concern, we consider a more general simulation setup in which asset returns are driven by their PCs and the conditional mean of consumption growth is determined by large or small PCs or even their linear combinations.

Specifically, we assume the following multifactor model for asset returns $\mathbf{r}_t^e$:

$$\mathbf{r}_t^e = \boldsymbol{\mu}_r + \boldsymbol{\rho}^r \mathbf{u}_t + \mathbf{w}_t^r, \quad \mathbf{u}_t \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}_K, \mathbf{I}_K), \quad (\boldsymbol{\rho}^r)' \boldsymbol{\rho}^r = \mathbf{I}_K, \quad \mathbf{w}_t^r \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}_N, \boldsymbol{\Sigma}_r), \quad (7)$$

where $\mathbf{u}_t$ denotes pseudo-true latent factors, $\boldsymbol{\rho}^r$ are their factor loadings, $\boldsymbol{\mu}_r$ is the vector of unconditional mean returns, and $\mathbf{w}_t^r$ are idiosyncratic shocks with a diagonal covariance matrix. $\Delta c_{t-1,t}$ is simulated using equation (2).

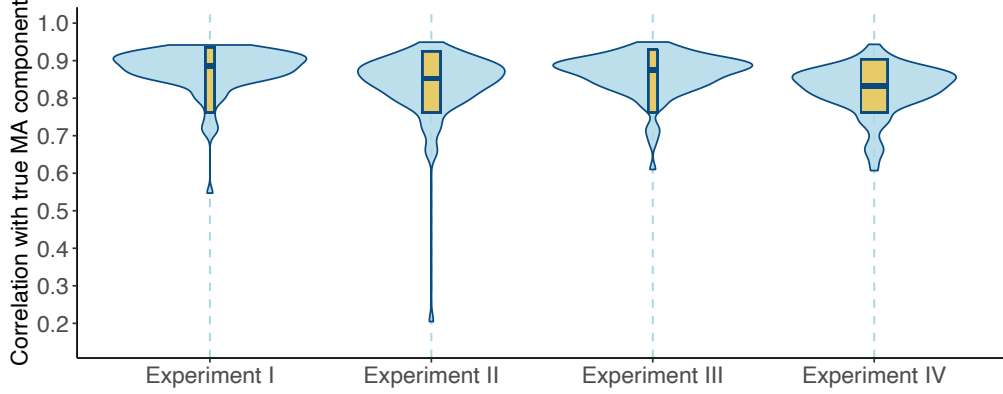
We consider four scenarios: (I) $\boldsymbol{\rho}^r$ contains the eigenvectors of the first and second PCs of

returns ($K = 2$), and $f_t = u_{1t}$, (II) $\boldsymbol{\rho}^r$ contains the eigenvectors of the first and second PCs of returns ($K = 2$), and $f_t = u_{2t}$, (III) $\boldsymbol{\rho}^r$ contains the eigenvectors of the first and second PCs of returns ($K = 2$), and $f_t = (u_{1t} + u_{2t})/\sqrt{2}$, and (IV) $\boldsymbol{\rho}^r$ contains the eigenvectors of the first five PCs ($K = 5$), and $f_t = (u_{1t} + u_{4t})/\sqrt{2}$. In addition, we estimate the eigenstructure of returns using observed return data of 37 stock portfolios and calibrate them such that $(\boldsymbol{\rho}_1^r)' \boldsymbol{\rho}_1^r = 16(\boldsymbol{\rho}_2^r)' \boldsymbol{\rho}_2^r = 25(\boldsymbol{\rho}_3^r)' \boldsymbol{\rho}_3^r = 36(\boldsymbol{\rho}_4^r)' \boldsymbol{\rho}_4^r = 144(\boldsymbol{\rho}_5^r)' \boldsymbol{\rho}_5^r$. Hence, the first latent factor is a dominant one. In all simulations, we calibrate $\boldsymbol{\rho}^c$ using the estimates obtained in Figure 3, and the variance of w_{ct} such that the time-series R^2 explained by the MA component of f_t is 23–24% in the consumption equation, consistent with our empirical evidence presented later.

Figure 2 presents the distribution of correlations between the estimated MA components and their pseudo-true values, that is, $\text{Corr}(\sum_{j=0}^{\bar{S}} \rho_j^c f_{t-j}, \sum_{j=0}^{\bar{S}} \hat{\rho}_j^c \hat{f}_{t-j})$. If we perfectly recover the MA component, the correlation coefficients should equal 1. It is worth noting that the identification of $\sum_{j=0}^{\bar{S}} \rho_j^c f_{t-j}$ depends on whether (1) we recover f_t and (2) we precisely estimate ρ_j^c . In experiment I, f_t is the strong factor (the first PC of asset returns), so identifying f_t is relatively simple. The box plot confirms this: Our estimates identify virtually the entire MA component in consumption growth, with a mean correlation coefficient of 0.89 (and a 90% confidence interval of [0.76, 0.93]), while the correlation between the true and filtered f shocks is 98%.

Moreover, our identification strategy does not mechanically recover the first principal component of $\mathbf{r}_t^e$. Experiment II assumes that f_t is the second PC of asset returns, whose variance is $\frac{1}{16}$ of the first PC, meaning that it drives on average only 4.4% of the time series variation of returns. However, our method can still recover the MA component in this challenging case. In particular, the second column of Figure 2 shows that the mean correlation is about 0.85, while the mean correlation between the true and filtered shocks is 93%, quite close to the value of experiment I. This implies that in this experiment, in a sample of the same size as the historical one, the shocks recovered by our method are not contaminated by the first principal component in a meaningful way. That is, our method does not deliver filtered consumption shocks that mechanically explain a large part of the time series variation of asset returns.

Figure 2: Correlation between estimated and pseudo-true MA components.



The graph presents the distribution of correlations between the estimated MA components and their pseudo-true values, that is, $\text{Corr}(\sum_{j=0}^{\bar{S}} \rho_j^e f_{t-j}, \sum_{j=0}^{\bar{S}} \hat{\rho}_j^e \hat{f}_{t-j})$. The model is estimated under the assumption that $\bar{S} = 14$. The simulated time-series sample size is 200 (quarters), under the following four simulation setups:

Experiment I: ρ^r contains the eigenvectors of the first two PCs of asset returns ($K = 2$), and $f_t = u_{1t}$.

Experiment II: ρ^r contains the eigenvectors of the first two PCs of asset returns ($K = 2$), and $f_t = u_{2t}$.

Experiment III: ρ^r contains the eigenvectors of the first two PCs ($K = 2$), and $f_t = (u_{1t} + u_{2t})/\sqrt{2}$.

Experiment IV: ρ^r contains the eigenvectors of the first five PCs ($K = 5$), and $f_t = (u_{1t} + u_{4t})/\sqrt{2}$.

Another concern of our approach is the single-factor structure in the consumption growth equation; that is, we allow for only one f_t . To address this critique, experiment III assumes that the true f_t is a linear combination of the top two PCs of $\mathbf{r}_t^e$. The simulation results are assuring: The mean correlation between the estimated and true MA components is 0.88, and the 90% confidence interval is similar to that of experiment I (while the mean correlation between true and filtered MA shocks is 96%).

Finally, the fourth experiment simulates returns using a five-factor model, and f_t consists of the first and fourth PCs. Despite the challenge that this scenario presents, we are still able to identify the MA component with a similar correlation coefficient as in the other three experiments (and a mean correlation of filtered MA shocks with the true ones of about 92%). However, the confidence interval is wider than that of experiment III, in which we consider a two-factor model. As shown in later sections, adding more latent factors in returns only increases estimation noise, thus leading to wider confidence intervals.

Overall, there are three main takeaways from the simulation study. First, our method does not mechanically identify the first principal component of asset returns as the driver

of the conditional mean shocks. Second, we can identify f_t even when it is a small principal component or a linear combination of several latent factors of asset returns. That is, the filtered MA innovations delivered by our method do not mechanically explain a large share of the time series variation of asset returns. Finally, it is essential to consider both single-factor and multifactor models. If we do not identify additional valuable information in consumption growth using multifactor models, the simple single-factor model is preferred because we can achieve sharper estimates and statistical inference.

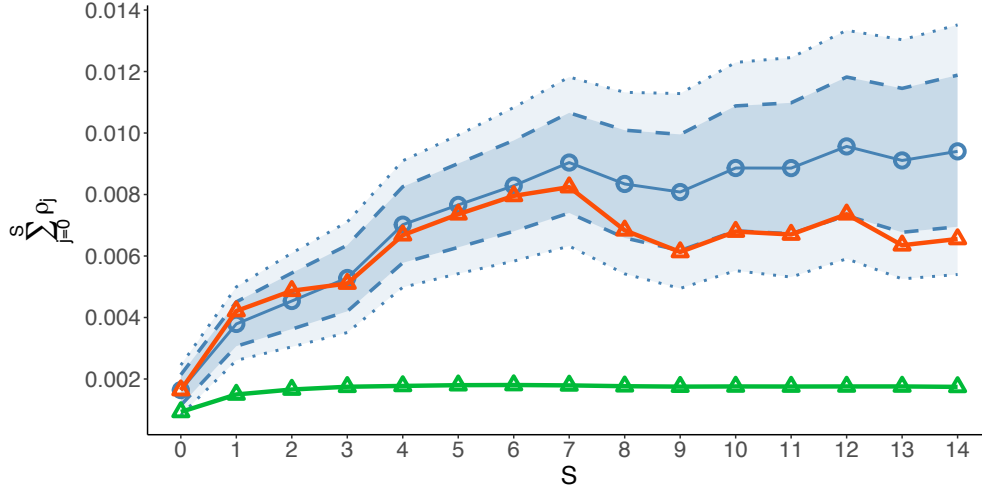
III Empirical Evidence

In this section, we bring the state-space formulation in equations (2)–(3), as well as its generalization in equations (5)–(6), to the data to recover the conditional consumption mean process (subsections III.1, III.3–III.5). Furthermore, our analysis identifies the forms of volatility processes that are consistent with the data, identifies the degree of commonality in the consumption and returns stochastic volatility processes, and determines whether they drive time-varying risk premia (subsections III.2 and III.6).

We first focus on the conditional consumption growth mean dynamics in subsections III.1, its implications for consumption volatility dynamics (subsection III.2), and the loadings of asset returns on the shocks spanned by consumption (subsection III.3). In these subsections, we focus on the most robust specification in that we consider only one latent factor (f), do not include external predictors, leave the covariance structure of returns unrestricted, and do not impose a particular dynamic on the volatility processes. We begin with this streamlined formulation because it is consistent for the estimation of the conditional mean process of consumption and its forecast errors under general conditions and, hence, it allows us to analyze the properties of consumption volatility without taking a stand on its true process (see, e.g., Engle (1982)).

We then confirm that the results are stable if we consider external predictors for consumption and asset returns (subsection III.4) and multiple latent factors driving asset returns (subsection III.5), if we explicitly model stochastic volatility of all shocks (subsection III.6), and if we consider different MA lengths and cross-sections of asset returns (subsection III.7).

Figure 3: Cumulative impulse response function of consumption growth to a one-standard-deviation shock spanned by asset returns.



Posterior means of the cumulative response function of consumption growth (solid line with circles), with the centered posterior 90% (dotted lines) and 68% (dashed lines) coverage regions. Red line with triangles denotes the first principal component of $cov(r_{i,t}^{ex}, \Delta c_{t,t+1+S})$. Quarterly data, 1963:Q3-2019:Q4. Green line with triangles is the simulated cumulative impulse responses assuming that (1) monthly consumption growth is independent over time and has a contemporaneous correlation of 0.20 with the monthly f_t shock, and (2) monthly consumption data are benchmarked to the annual data (see Internet Appendix H for details).

We also show that the key stylized facts we uncover in the data are confirmed nonparametrically without the additional assumptions needed for the state-space model estimation.

III.1 The consumption mean process

The consumption growth representation in equations (2) and (5) is similar to the MA decomposition and allows us to infer the dynamics of multi-period consumption growth ($\Delta c_{t,t+1+S}$) in response to a common and/or bond-specific shock. Figure 3 depicts the (cumulated) loadings of consumption on the latent factor f as a function of the horizon S . At $S = 0$, as in the case of a standard consumption-based asset pricing model, the MA component of consumption virtually does not load on the common factor. Instead, as S increases, the impact of the common factor becomes increasingly pronounced, leveling off at approximately $S = 10$. At this horizon, the effect is economically substantial: The cumulative response of consumption growth to a one-standard-deviation shock is about 1%. Nevertheless, this large effect is *not* excessive: As shown in Internet Appendix E, it does not violate the bound on the long-run standard deviation of consumption growth obtained in Dew-Becker (2017).

Importantly, the finding in Figure 3 reconciles seemingly discordant empirical claims in the literature. For instance, Kleibergen and Zhan (2020) argue that the correlation between stock returns and consumption growth is small and not significantly different from zero. In contrast, Parker and Julliard (2005) find a large and significant long-run response of consumption to asset return shocks. Our MA representation sheds light on this apparent contradiction by pinning down the slow propagation mechanism of shocks spanned by financial markets into consumption: It is the same consumption shock that is both weakly identified in the short horizon yet strong and evident at business cycle frequency. Furthermore, the tight confidence intervals in the figure emphasize that our parametric state-space approach has a higher power of detecting the link between financial markets and consumption compared to the nonparametric approaches in the previous literature that yield high sampling uncertainty.

Note that allowing for a bond-specific latent factor (equations (5)–(6)) leaves the consumption loadings on f shocks virtually unchanged, and consumption does not significantly load on the bond-specific factor b_t . See, respectively, Figure IA.21 in Internet Appendix P.

The relatively tight restrictions on the parametric model in equations (2)–(3) allow us to pin down the parameters of the joint consumption–returns process with a high degree of precision. However, this comes at the price of imposing (weak) constraints on the data-generating process, some of which may, in principle, not hold in the data. However, the strongest prediction of our parametric setting – the term structure of asset exposure to consumption risk – can be tested without all the ancillary assumptions of the state-space model.

To see this, note that equations (2)–(3) imply a particular pattern in the covariances of asset returns with multi-period consumption growth; that is, for any asset i ,

$$\begin{aligned} cov(r_{i,t+1}^{ex}, \Delta c_{t,t+1}) &= \rho_i^r \rho_0, \\ cov(r_{i,t+1}^{ex}, \Delta c_{t,t+2}) &= \rho_i^r (\rho_0 + \rho_1), \quad \text{and} \\ &\dots \\ cov(r_{i,t+1}^{ex}, \Delta c_{t,t+k}) &= \rho_i^r \left(\sum_{j=1}^k \rho_{k-j} \right). \end{aligned} \tag{8}$$

Therefore, the term structure of asset exposures to consumption risk is driven by a single

common component: $(\rho_0, \rho_0 + \rho_1, \dots, \sum_{j=1}^k \rho_{k-1})'$, that is, the cumulative response function of consumption to an f_t shock. This property is not affected by the potential presence of cross-sectional correlations between stocks and bonds or additional factors driving stocks and bonds that are orthogonal to consumption. Therefore, if the time-varying dynamics of consumption growth in equation (2) describes well the data-generating process, we should be able to recover the same pattern of loadings by simply extracting the first uncentered principal component of $cov(r_{i,t+1}^{ex}, \Delta c_{t,t+k})$ at different horizons k .

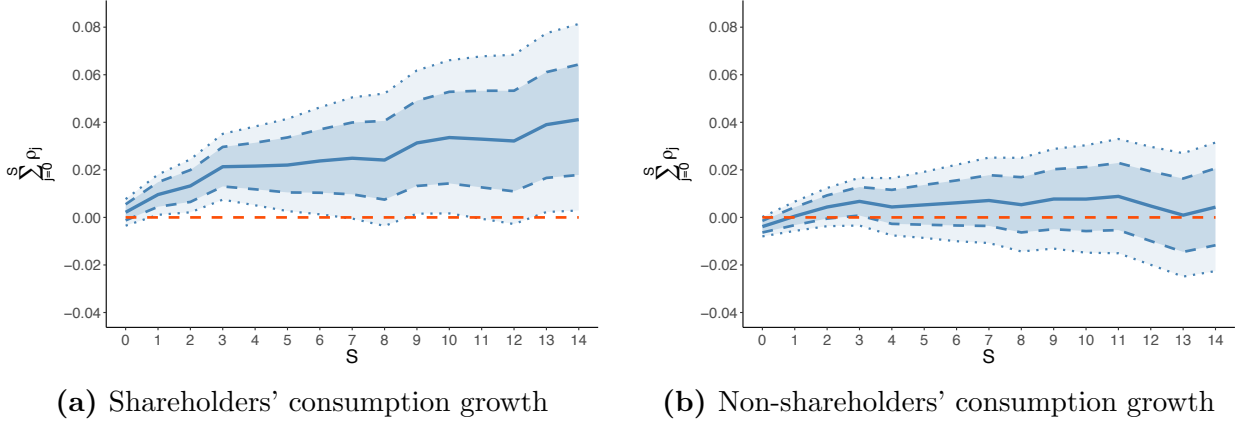
The red line with triangles in Figure 3 illustrates our findings. Remarkably, the loadings on the first PC of consumption term structure exposure almost exactly match the cumulated response function from the state-space model, therefore identifying the same persistent time-varying mean for consumption growth.⁸

Recall that the identifying restriction for the finding above is that, for a representative agent, the intertemporal Euler equation determining equilibrium prices implies that both returns and consumption react to a common set of innovations (e.g., wealth shocks and changes in expectations). But is this really what we uncover? For an agent who does not participate in the equity market, such an Euler equation argument does not hold. Hence, we can verify this mechanism by using micro-data on the consumption of stockholders and non-stockholders.

Figure 4 uses the consumption series constructed in Malloy, Moskowitz, and Vissing-Jorgensen (2009) over the sample of 1982:Q1 to 2004:Q3 and reestimates our state-space model separately for these two types of households. Although the confidence bands are wider than in our baseline estimation due to the short sample size, the message is clear: Consumption growth of households that participate in the financial market (left panel) reacts significantly, and gradually over multiple quarters, to shocks spanned by financial assets, whereas the consumption of other households (right panel) does not. This further confirms the power and the economic mechanism underlying our identification strategy: Some of the shocks affecting the households that participate in financial markets are reflected in *both* equilibrium asset prices and consumption. Hence, we use this insight to uncover their joint dynamics and compare them with those postulated in popular macro-finance models. Note also that the estimated effect of a one-standard-deviation shock for shareholders' consumption is actually larger, by a factor of 2–3, than the effect uncovered in aggregate data –

⁸In the figure, the level of the first PC is normalized to have the same origin as the ρ_0 estimated from the state-space formulation.

Figure 4: Cumulative impulse response functions of shareholders’ and non-shareholders’ consumption growth to a one-standard-deviation shock spanned by asset returns.



Posterior means of the cumulative response functions of consumption growth, centered posterior 90% (dotted lines) and 68% (dashed lines) coverage regions. Panels (a) and (b) use shareholders and non-shareholders’ consumption growth, respectively, following Malloy, Moskowitz, and Vissing-Jorgensen (2009). Data are downloaded from Tobias Moskowitz’s website. We use the “Dn1” and “Ds1” variables as proxies for non-shareholders’ and shareholders’ quarterly consumption growth. Data sample: 1982:Q1 to 2004:Q3.

exactly as one would expect, given that the aggregate series also contains the consumption of non-shareholders.

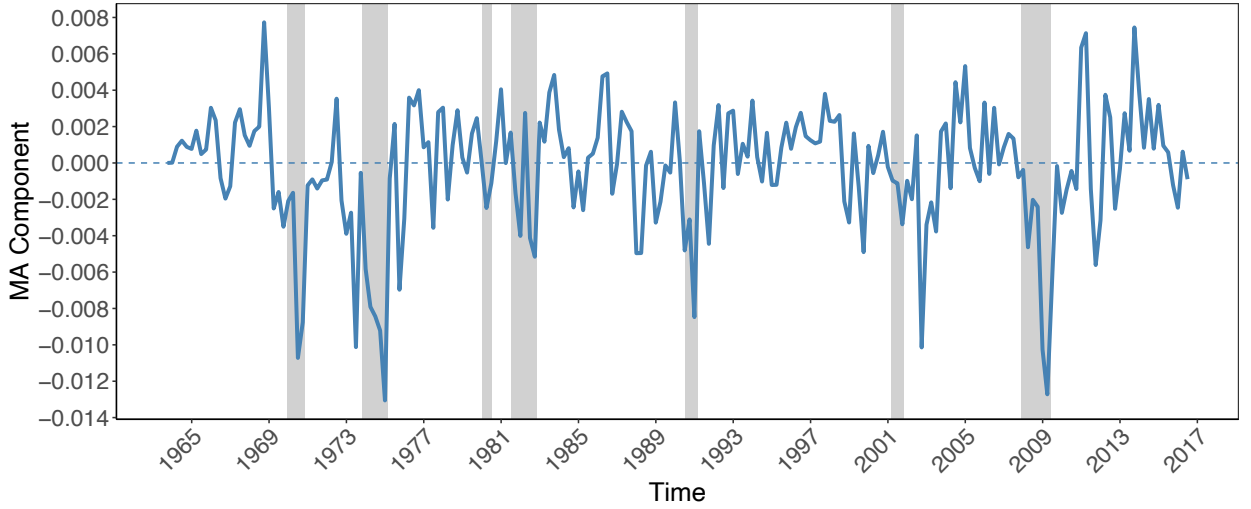
Figure 5 shows the (posterior mean of the) MA component of consumption, based on our filtered f_t innovations. The slow-moving component within consumption aptly captures the business-cycle fluctuations and has a pronounced exposure to recession risk. Furthermore, this component generates economically large swings in *quarterly* consumption growth, with contractions and expansions of about 1% being not uncommon.

III.1.1 Measurement issues of consumption

There are two possible measurement issues in quarterly data that might bias our estimates in Figure 3: time averaging and benchmarking of the quarterly consumption series. We now turn to evaluate their potential impact.

The evidence of persistency of the conditional consumption mean process that we document in Figure 3 might be contaminated by the well-known time-aggregation “bias” (see, e.g., Breeden, Gibbons, and Litzenberger (1989)). That is, consumption growth measured from quarter t to $t + 1$ is in fact a moving average of true consumption growth from $t - 1$ to $t + 1$ (as shown in Equation (IA.11) of the Internet Appendix). We formally address this

Figure 5: Moving average component of consumption growth.



Posterior mean of the moving average f_t component of consumption growth. Grey areas denote NBER recessions. Estimate based on the single-factor model in equations (2)–(3), with $\bar{S} = 14$. The cross-section of base assets includes 25 size- and value-sorted portfolios, 12 industry portfolios, and nine bond portfolios.

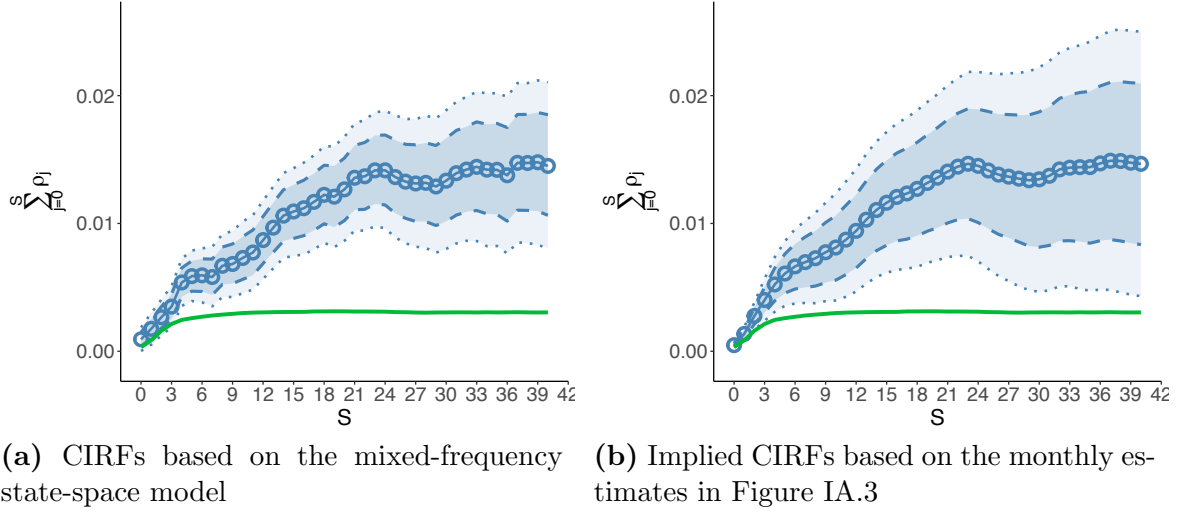
issue in two ways.

First, in Section G of the Internet Appendix, we introduce a mixed-frequency state-space formulation that can be overlayed on our estimation framework to directly model the time aggregation of quarterly consumption. Therein, we also show that, under the null of serially uncorrelated monthly consumption growth, we should not observe significant impulse responses of consumption after two quarters, even in the presence of time-aggregation bias. Furthermore, we estimate a latent factor model for monthly excess returns and allow quarterly consumption growth (observed only at the end of each quarter) to react slowly to the monthly asset return shocks as in Equation (2).

Second, as an additional way of addressing the time-aggregation bias, in Section F of the Internet Appendix, we reestimate our dynamic model for consumption and asset returns using monthly consumption and return data. Note that monthly consumption requires considering the well-known issue of measurement error in these data, which is directly modeled in our extended state-space formulation (see equation (IA.8) of the Internet Appendix).

Figure 6 presents the cumulative responses of quarterly consumption growth to one-standard-deviation monthly asset return shocks. We report the mixed-frequency state-space estimates in Panel (a), where the impulse response of consumption continues to increase until about 24 months, with a cumulative effect of roughly 0.015. Panel (b), instead, depicts the cumulative impulse function obtained with the monthly consumption data and mapped

Figure 6: Cumulative impulse response function of quarterly consumption growth to a one-standard-deviation shock spanned by monthly asset returns.



Posterior means of the cumulative impulse response function of quarterly consumption growth (solid lines with circles) to monthly asset return shocks, with centered posterior 90% (dotted lines) and 68% (dashed lines) coverage regions. The x-axis denotes the number of *monthly* lags. Panel A uses the mixed-frequency state-space model in Section G of the Internet Appendix. Panel B is based on the impulse responses of monthly consumption growth to monthly asset return shocks, converted into the CIRFs of quarterly consumption growth to monthly asset return shocks (see equation (IA.15) of the Internet Appendix). Green lines display the simulated cumulative impulse responses, assuming that (1) monthly consumption growth is independent over time and has a contemporaneous correlation of 0.20 with the monthly f_t shock, and (2) monthly consumption data are benchmarked to the annual data (see Internet Appendix H for details).

into quarterly consumption responses. The estimates in Panels (a) and (b) are strikingly similar, with wider confidence intervals in the case of monthly consumption data (due to the substantial measurement error). Furthermore, as Figure IA.3 of the Internet Appendix shows, monthly consumption growth still slowly adjusts to asset return shocks: The f_t shocks explain more than 11% of the time-series variation of *monthly* consumption growth.

But how would the time-aggregated quarterly consumption growth react to monthly asset return shocks if there were no predictability in monthly consumption? We answer this question in Figure IA.11 of the Internet Appendix by conducting a counterfactual exercise. Specifically, we impose that monthly consumption growth correlates only with contemporaneous asset return shocks. This implies that the time-aggregated quarterly consumption growth comoves with only contemporaneous and lagged asset returns up to four months.⁹ In this setting, we find a much smaller cumulative impulse response of quarterly consumption growth that peaks at less than 0.004 (one quarter of what we show in Figure 6). Hence,

⁹That is, we impose $\bar{S} = 0$ in equation (IA.14) of the Internet Appendix.

the time aggregation of consumption is unlikely to fully explain the predictability that we observe in quarterly consumption growth.

In addition to time aggregation, the serial dependence of quarterly consumption growth could be biased by benchmarking (see Wilcox (1992) and Triplett (1997)). Specifically, monthly and quarterly estimates of consumption are based on the Monthly Retail Trade Survey, which is of lower quality than the annual survey. Because the four quarters of consumption never precisely match the annual measure, the monthly and quarterly estimates are ex-post revised to benchmark to the annual estimates.

What would our state-space estimation method yield if consumption were IID but time-aggregated and benchmarked? We answer this question by simulation (see details in Internet Appendix H). In particular, we add to Figures 3 and 6 the simulated CIRFs (green lines), assuming that i) monthly consumption growth is independent over time and has a contemporaneous correlation of 0.20 with the monthly f_t shock¹⁰ and ii) monthly consumption data are benchmarked to the annual consumption data. Two observations are in order.

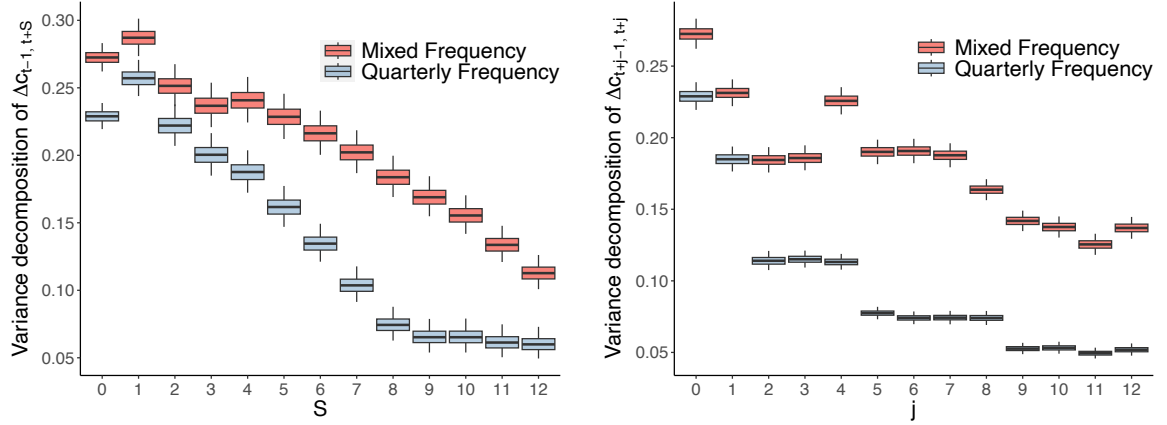
First, even if the true monthly consumption growth were serially independent, we would still detect slow responses of quarterly consumption growth to asset return shocks due to time aggregation and benchmarking. Furthermore, the asset return shock would predict quarterly consumption growth up to four quarters ahead. However, the cumulative impulse responses would be flat after the benchmarking horizon of four quarters.

Second, nevertheless, time aggregation and benchmarking of consumption data cannot fully explain the predictability of consumption growth that we estimate using our state-space formulation. In the observed data, as shown in Figures 3 and 6, the cumulative responses increase until the seventh quarter and are much larger than those predicted by time-aggregation bias and benchmarking (the green lines in the figures). Even when calibrating a much larger contemporaneous correlation between monthly consumption growth and f_t (e.g., 0.4 in Figure IA.15 of the Internet Appendix), our state-space model estimates of the cumulative responses are much more persistent and sizable than those obtained in IID time-aggregated and benchmarked consumption data.

Overall, the large degree of consumption predictability that we uncover cannot be rationalized by time aggregation and benchmarking of consumption data.

¹⁰In the real data, this contemporaneous correlation is about 0.17. Figure IA.15 of the Internet Appendix explores different correlation coefficients of between 0.10 and 0.40.

Figure 7: Share of consumption growth variance driven by its moving average component.



Box plots (posterior 95% coverage area) of the percentage of time-series variances of consumption growth explained by the MA component. The state-space models are estimated at both quarterly (blue bars) and mixed (red bars) frequencies. Plots report the unadjusted R-squared. Left panel: Cumulated consumption growth $\Delta c_{t-1,t+S}$. Right panel: One-period consumption growth $\Delta c_{t-1+j,t+j}$. Estimates based on single-factor model, with $\bar{S} = 14$. The cross-section of base assets includes 25 size- and value-sorted portfolios, 12 industry portfolios, and nine bond portfolios.

III.1.2 Predicting consumption

But how much of the total consumption volatility can this slow-moving component explain? The answer is more than a quarter, which is large (and sharply estimated) compared to the leading asset pricing frameworks: For example, this is more than twice as large as in the long-run risk framework of Bansal and Yaron (2004).¹¹ Figure 7 demonstrates that the common factor is responsible for roughly 23% of the variation in the one-period nondurable consumption growth, 26% of the two-period consumption growth, and so on, followed by a slow decline toward slightly more than 6% for the 12-period growth.¹² Furthermore, as shown in the figure, the predictability of consumption is even *higher* at the longer horizons when we estimate the mixed-frequency state-space model to correct the time aggregation. In this case, the MA component can explain about 12% of the quarterly consumption growth at the three-year horizon. Interestingly, the model retains significant predictive power (albeit lower), even for the one-period consumption three years ahead.

But does the conditional mean component uncovered by our method do a good job at predicting consumption? Does it survive horse races with the canonical predictors? To

¹¹This quantity is zero in frameworks that, for parsimony, do not model any predictability in consumption (e.g., Campbell and Cochrane (1999) and Barro (2006)).

¹²As shown in Figure IA.22 of the Internet Appendix, adding a bond-specific factor has a minimum impact on the explanatory power of the model for future consumption growth.

answer these questions, we explore whether the future realized consumption growth at short to medium horizons could be explained by the estimated conditional mean implied by our MA model. Specifically, given the estimated MA coefficients $\{\hat{\rho}_j\}_{j=0}^{\bar{S}}$, we can estimate the conditional mean of future one-period consumption growth as

$$\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}] = \mu_c + \sum_{s=j}^{\bar{S}} \hat{\rho}_s f_{t+j-s}, \quad (9)$$

where j ranges from 1 to 12 quarters. To avoid forward-looking bias and preserve consistency of our predictive regressions, we employ a (one-side) Kalman filter (rather than a Kalman smoother) to extract the latent state f_t . We next regress $\Delta c_{t+j-1,t+j}$ on $\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$,

$$\Delta c_{t+j-1,t+j} = \beta_{0,j} + \beta_{1,j} \hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}] + \beta_{2,j} \text{controls}_t + e_{t+j-1,t+j}^c, \quad (10)$$

and test whether the slope coefficient, $\beta_{1,j}$, is significantly different from one—the value that we should obtain in a long sample with a correctly specified model. As we do not reestimate the MA coefficients to fit future consumption growth, this exercise is an additional test of the validity of our method.

Table 2 reports the predictive regression results. In Panel A, we regress future (one-period) consumption growth (from one to 12 quarters ahead) on our estimated conditional means. The predictive R^2 s are extremely close to those obtained in Panel B of Figure 7, with the predictability after two years declining faster than shown in the figure. This small decline in predictability is not surprising, given the one-sided nature of the Kalman filter compared with the Kalman smoother in the figure. Moreover, the slope coefficients associated with the conditional mean are extremely close to, and statistically indistinguishable from, one—as should be under the null of our specification. As a benchmark, in Table IA.III of the Internet Appendix, we repeat the same predictability analysis using as a proxy for the conditional mean the VAR-based estimate of Parker (2001). At the rather short horizon (1–2 quarters), the VAR has performance similar to our state-space formulation, but at longer horizons, its performance quickly deteriorates, with R^2 s essentially equal to zero and slope coefficients much smaller than one after one year.

Panels B to D of Table 2 conduct horse races with the canonical predictors: Lagged consumption and GDP growth, as well as a Kalman filter estimate from a standard AR(1)

latent expected growth model for consumption growth.¹³ The results therein confirm that our conditional estimate captures information not fully spanned by these canonical predictors. First, the slope coefficient associated with the conditional mean is once again close to and indistinguishable from one. Second, the (unadjusted) R^2 s are only marginally higher than those in Panel A, highlighting the fact that most of the information contained in these variables is already captured by our conditional consumption mean process. Third, the added predictors never drive out the statistical significance of our conditional mean, and are themselves statistically different from zero only in rare cases (3 out of 36, i.e., at the expected Type I error frequency) and only at the short horizon (one or two quarters).

As additional horse races, in Tables IA.VII and IA.VIII of the Internet Appendix, we consider the mean consumption forecasts from the Survey of Professional Forecasters (SPF) as well as the news-based index of Liu and Matthies (2022).¹⁴

The SPF forecasts are available for only one- to four-quarter horizons and over a much shorter sample than our baseline (starting from 1981:Q3). Panel A of Table IA.VII reveals that the SPF predictive ability is much lower (about half in most cases) than the one of our conditional mean estimates in Panel A of Table 2. Furthermore, when used in conjunction with our conditional mean (Panel B of Table IA.VII), the SPF forecasts do not drive out the statistical significance of the former (albeit at the one-quarter horizon, but not others, the coefficient is statistically smaller than one). Moreover, SPF forecasts only marginally improve the predictive power achievable with our conditional mean model. Interestingly, while the predictive ability of SPF forecasts three and four quarters ahead is small (see Panel A of Table IA.VII), the one of our conditional mean model is substantial, with predictive R^2 s of about 10% (see Panel A of Table 2). The robustness of our conditional model to the inclusion of SPF forecasts is remarkable since professional forecasters are likely to use the information in financial markets to predict future consumption growth.

As shown in Panel A of Table IA.VIII, after controlling for our MA-implied conditional consumption mean, the news index of Liu and Matthies (2022) has no incremental predictive

¹³ Specifically, we consider the Kalman filter estimate from an AR(1) latent expected growth model:

$$\Delta c_{t-1,t} = \mu_c + x_t + w_t^c, \quad x_t = \rho_x x_{t-1} + w_t^x, \quad w_t^c \sim \mathcal{N}(0, \sigma_{wc}^2), \quad w_t^x \sim \mathcal{N}(0, \sigma_{wx}^2), \quad (11)$$

which implies that the latent state x_t should predict future consumption growth.

¹⁴Two caveats are in order. First, the SPF forecasts are based on total real personal consumption (PCE). Second, Liu and Matthies (2022) show that their news index has forecasting power for nondurable *plus* service consumption. Our baseline consumption series is the nondurable one instead.

Table 2: Validating the predictability of the conditional consumption mean

$j =$	1	2	3	4	5	6	7	8	9	10	11	12
Panel A. Regress $\Delta c_{t+j-1,t+j}$ on $\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$												
$\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$	0.998	0.964	0.964	0.965	0.932	0.913	0.963	1.129	1.021	0.984	1.075	1.080
s.e. (OLS)	(0.146)	(0.196)	(0.202)	(0.204)	(0.271)	(0.287)	(0.313)	(0.353)	(0.409)	(0.423)	(0.561)	(0.561)
s.e. (NW, lag=12)	(0.196)	(0.199)	(0.195)	(0.203)	(0.190)	(0.210)	(0.182)	(0.204)	(0.330)	(0.374)	(0.461)	(0.461)
Predictive R^2	0.182	0.104	0.098	0.097	0.054	0.046	0.043	0.047	0.029	0.025	0.017	0.017
Panel B. Additional control: Lagged consumption growth												
$\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$	0.932	0.896	0.867	0.947	1.007	0.894	0.963	1.100	1.024	1.028	1.032	1.087
s.e. (OLS)	(0.151)	(0.200)	(0.210)	(0.208)	(0.278)	(0.295)	(0.314)	(0.350)	(0.411)	(0.435)	(0.572)	(0.563)
s.e. (NW, lag=12)	(0.192)	(0.190)	(0.213)	(0.222)	(0.241)	(0.224)	(0.185)	(0.226)	(0.325)	(0.376)	(0.453)	(0.466)
Δc_t	0.107	0.103	0.110	0.032	-0.079	0.019	-0.024	-0.150	-0.006	-0.030	0.026	-0.017
s.e. (OLS)	(0.064)	(0.066)	(0.067)	(0.066)	(0.068)	(0.068)	(0.067)	(0.065)	(0.064)	(0.066)	(0.066)	(0.065)
s.e. (NW, lag=12)	(0.067)	(0.074)	(0.074)	(0.066)	(0.111)	(0.075)	(0.058)	(0.087)	(0.057)	(0.059)	(0.065)	(0.061)
Predictive R^2	0.193	0.114	0.110	0.098	0.060	0.047	0.044	0.070	0.029	0.026	0.018	0.018
Panel C. Additional control: Lagged GDP growth												
$\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$	0.839	0.781	0.907	1.023	0.928	0.942	0.957	1.173	1.051	0.994	1.097	1.107
s.e. (OLS)	(0.149)	(0.211)	(0.210)	(0.208)	(0.276)	(0.287)	(0.313)	(0.353)	(0.415)	(0.436)	(0.563)	(0.563)
s.e. (NW, lag=12)	(0.207)	(0.223)	(0.202)	(0.224)	(0.192)	(0.211)	(0.186)	(0.198)	(0.331)	(0.342)	(0.436)	(0.491)
ΔGDP_t	0.199	0.136	0.061	-0.079	0.004	-0.072	-0.061	-0.094	-0.025	-0.005	-0.028	-0.045
s.e. (OLS)	(0.056)	(0.062)	(0.059)	(0.058)	(0.059)	(0.059)	(0.059)	(0.058)	(0.057)	(0.058)	(0.057)	(0.057)
s.e. (NW, lag=12)	(0.053)	(0.052)	(0.050)	(0.072)	(0.063)	(0.055)	(0.074)	(0.080)	(0.052)	(0.052)	(0.064)	(0.074)
Predictive R^2	0.229	0.124	0.103	0.105	0.054	0.053	0.048	0.058	0.030	0.025	0.018	0.020
Panel D. Additional control: Latent state (x_t) of AR(1) expected growth model												
$\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$	0.901	0.853	0.866	0.963	1.010	0.919	0.965	1.141	1.033	1.014	1.050	1.086
s.e. (OLS)	(0.152)	(0.204)	(0.213)	(0.210)	(0.281)	(0.293)	(0.313)	(0.351)	(0.415)	(0.436)	(0.571)	(0.563)
s.e. (NW, lag=12)	(0.197)	(0.195)	(0.227)	(0.234)	(0.253)	(0.222)	(0.187)	(0.218)	(0.322)	(0.374)	(0.460)	(0.468)
x_t	0.299	0.272	0.213	0.006	-0.155	-0.014	-0.141	-0.285	-0.027	-0.042	0.035	-0.034
s.e. (OLS)	(0.141)	(0.147)	(0.148)	(0.144)	(0.148)	(0.147)	(0.144)	(0.142)	(0.141)	(0.143)	(0.142)	(0.140)
s.e. (NW, lag=12)	(0.135)	(0.158)	(0.163)	(0.166)	(0.265)	(0.176)	(0.149)	(0.199)	(0.129)	(0.131)	(0.149)	(0.146)
Predictive R^2	0.200	0.118	0.107	0.097	0.059	0.046	0.048	0.065	0.029	0.026	0.018	0.018

The table summarizes the regressions in which future realized growth rates of nondurable consumption ($\Delta c_{t+j-1,t+j}$, $1 \leq j \leq 12$) are forecasted by several predictors. In Panel A, we regress $\Delta c_{t+j-1,t+j}$ on the conditional consumption mean ($\hat{\mathbb{E}}_t[\Delta c_{t+j-1,t+j}]$) implied by the MA model (see equation (9)). In Panels B–D, we include additional controls, such as time- t consumption growth, time- t GDP growth, and time- t latent state x_t from the AR(1) latent expected growth model in equation (11). We report (1) the point estimates of the slope coefficients, (2) the OLS and Newey-West (12 lags) standard errors within the parentheses, and (3) the predictive R^2 . Sample: 1963:Q3–2019:Q4.

power for our baseline nondurable consumption at any horizon (from one to 12 quarters). Nevertheless, in Panel B, we confirm that the news index does help predict nondurable *plus* service consumption.

III.2 Clustering and predictability of total consumption volatility

Note that the estimation of the conditional mean of consumption growth in equations (2) and (5) is generally consistent even in the presence of time-varying consumption volatility. Therefore, the presence of volatility clustering can be assessed by analyzing the serial correlation of the squared one-step-ahead forecast errors (see, e.g., Engle (1982)) of the consumption growth process. This proxy for volatility has been used extensively in the

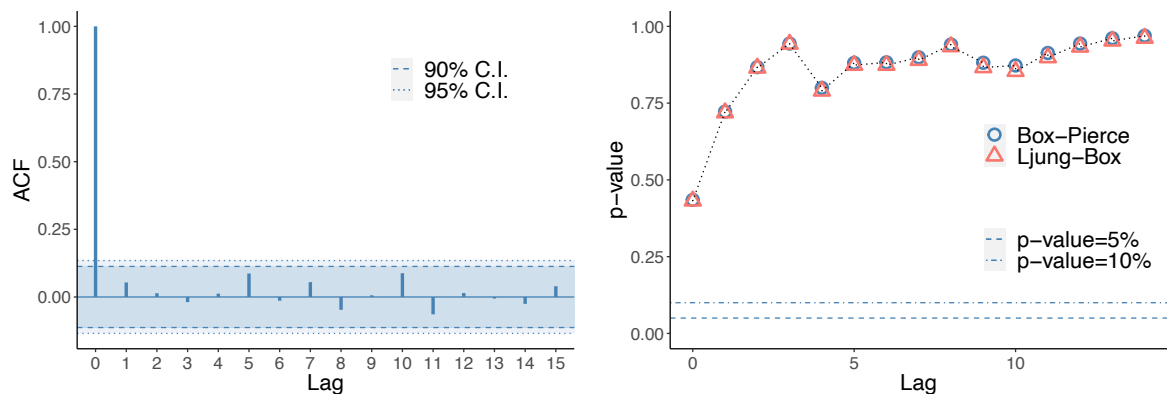
empirical consumption-based asset pricing literature, for example, Bansal, Khatchatrian, and Yaron (2005), Bansal, Kiku, and Yaron (2012), Beeler and Campbell (2012), and Chen (2017). That is, we examine the autocorrelation and predictability of $\widehat{Var}_t(\Delta c_{t,t+1}) := \left(\Delta c_{t,t+1} - \widehat{\mathbb{E}}_t[\Delta c_{t,t+1}]\right)^2$, where the conditional mean is computed at each t using the estimated ρ_j and θ_j coefficients and latent state variables $f_{\tau < t}$ and $b_{\tau < t}$.¹⁵ Note that this nonparametric volatility proxy does not distinguish the nature of the volatility being captured. For example, under the null of the consumption volatility process of Schorfheide, Song, and Yaron (2018), the square forecast errors would be a linear combination of the variances of innovations to immediate consumption growth (short-run news) and expected growth (long-run news). In Section III.6 we introduce a parametric model to be able to distinguish these potentially different volatility processes.

Figure 8 reports the autocorrelation function (left panel) as well as the p -values of the Ljung and Box (1978) and Box and Pierce (1970) tests (right panel) of joint significance of the autocorrelations of $\widehat{Var}_t(\Delta c_{t,t+1})$ and shows no evidence of volatility clustering in the consumption growth process. Nevertheless, conditional consumption volatility might still, in principle, be correlated with financial asset returns. We test this hypothesis by running linear predictive regressions of $\widehat{Var}_{t+h}(\Delta c_{t+h,t+h+1})$, at several horizons h , on the time- t first eight principal components of stock and bond returns – that is, we check whether asset returns can predict the nonparametric consumption volatility proxy. Note that, in line with the previous literature (see, e.g., Liew and Vassalou (2000)), this is the same test used to establish predictability of the first conditional moment of consumption growth, as shown in Appendix D. The p -values of the F -tests for these predictability regressions are depicted by the blue continuous line with circles in Figure 9. The p -values (which range from 0.2826 to 0.922) show that asset returns are not significant predictors of future consumption volatility.

We conduct two robustness checks of the above result. First, in Figures IA.9 and IA.10 of Internet Appendix G, we show that extremely similar results to those above are obtained in the mixed-frequency estimation setting, confirming that these are not due to a time-aggregation bias. Second, in Table IA.IX of the Internet Appendix, we run univariate regressions of $\widehat{Var}_t(\Delta c_{t,t+1})$ on persistent predictors that are possibly more informative about the level of volatility: the log price-dividend ratio of the market index as well as the measures of financial, macro, and real uncertainty of Jurado, Ludvigson, and Ng (2015) and

¹⁵ $\widehat{\mathbb{E}}_t[\Delta c_{t,t+1}]$ is the posterior mean of $\mu_c + \sum_{j=1}^{\bar{S}} \rho_j f_{t+1-j} + \sum_{j=1}^{\bar{S}} \theta_j b_{t+1-j}$ at each t .

Figure 8: Autocorrelation structure of consumption growth squared forecast errors.



Left panel: Autocorrelation function of $\widehat{Var}_t(\Delta c_{t,t+1})$ with 95% and 99% confidence bands. Right panel: p -values of Ljung and Box (1978) (red triangles) and Box and Pierce (1970) (blue circles) tests.

Ludvigson, Ma, and Ng (2021). The regression R^2 s are extremely small, ranging from 0% to 1.8%, and in only one case (that of the real uncertainty) we find a regressor to be statistically significant at canonical levels with one of the two types of standard errors that we consider.

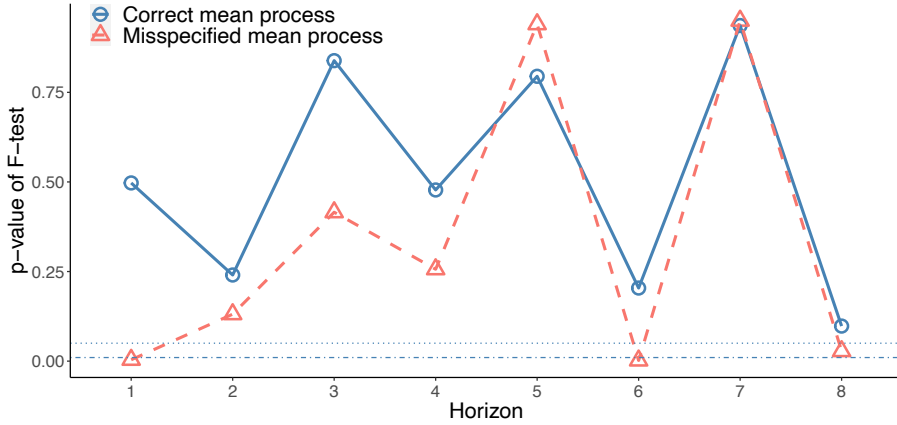
Since the clustering in total consumption volatility is weak at best, for this feature of the data to be revealed it is key to properly account for the conditional mean of the consumption process. Indeed, given our finding that a common latent factor drives both asset returns and consumption, if one were to erroneously model the conditional mean of consumption growth, one would be likely to find spurious evidence of volatility clustering. For instance, if one erroneously models the conditional mean of consumption as being constant, the autocorrelation of $\widehat{Var}_t(\Delta c_{t,t+1})$ would be mechanically different from zero. For example, the k -th autocorrelation of $(\Delta c_{t,t+1} - \mu_c)^2$, for $k \leq \bar{S}$, is proportional to

$$Cov \left(\left(\sum_{j=k}^{\bar{S}} \rho_j f_{t-j} + \theta_j b_{t-j} \right)^2, \left(\sum_{j=k}^{\bar{S}} \rho_{j-k} f_{t-j} + \theta_{j-k} b_{t-j} \right)^2 \right) \neq 0. \quad (12)$$

To verify that a misspecification of the consumption mean process leads to spurious evidence of time-varying volatility, we perform two exercises.

First, we again run predictability regressions of $\widehat{Var}_{t+h}(\Delta c_{t+h,t+h+1})$ on the first eight principal components of asset returns (the same predictive variables used in the preliminary evidence presented in Appendix D) but, as a misspecification benchmark, we construct these measures assuming a constant conditional mean for consumption growth. Summary statistics for these regressions are depicted by the dashed red line with triangles in Figure 9. The figure shows that the misspecification of the mean process generates spurious predictability

Figure 9: Predictability of consumption squared forecast errors.



Predictive regressions of $\widehat{Var}_{t+h}(\Delta c_{t+h,t+h+1})$ on the time- t first eight principal components of asset returns at several horizons h . We report the p -value of the F -test of joint significance of the covariates as well as the 5% and 1% significance thresholds (respectively, horizontal dot-dashed and dotted lines). The solid blue line with circles denotes statistics for the correctly specified conditional mean for the consumption growth process. The dashed red line with triangles corresponds to the assumption of a constant conditional mean.

of consumption volatility in three of the eight horizons considered. That is, modeling the mean of consumption growth without exploiting the information in asset returns and the flexibility of the MA representation leads to spurious evidence of time-varying volatility of consumption growth. Instead, with the robust MA specification of the mean process, there is no evidence of predictability in the volatility proxy of consumption growth (blue continuous line with circles of Figure 9).¹⁶

Second, in Table 3 we estimate ARCH(1), GARCH(1,1), and IGARCH(1,1) models¹⁷ for consumption volatility under different assumptions about the mean process. In Panel A, following the standard approach of Bansal, Khatchatrian, and Yaron (2005), Bansal, Kiku, and Yaron (2012), Beeler and Campbell (2012), Tédongap (2014), Chen (2017), and Zviadadze (2021), we identify the volatility process by postulating an AR(1) specification for the conditional mean of consumption growth. The AR(1)-GARCH(1,1) formulation in Panel A is the most often used in the literature to provide evidence of time-varying consumption volatility.¹⁸ In this case, there is statistically significant evidence of volatility clustering ($\beta > 0$ in Column (5)) with a half-life of volatility departures from the mean of about 5–6

¹⁶This is in line with the evidence of Dew-Becker and Giglio (2016, Appendix D) who, proxying consumption volatility with the realized volatility of the S&P500 index (as often assumed in the prior literature), find no predictability.

¹⁷See, respectively, Engle (1982), Bollerslev (1986), and Bollerslev and Engle (1986).

¹⁸In Section III.6 we provide an extensive analysis of SV processes in consumption and asset returns, of the type also commonly used in the literature.

Table 3: Estimates of ARCH(1), GARCH (1,1), and IGARCH(1,1) for different models

$$\Delta c_{t+1} = \mu_t + \epsilon_{t+1}$$

$$\sigma_{t+1}^2 = \omega + \alpha \epsilon_t^2 + \beta \sigma_t^2$$

	ARCH(1)		GARCH(1,1)			IGARCH(1,1)		
	ω	α	ω	α	β	ω	α	β
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: $\mu_t = \mu_0 + \mu_1^c \Delta c_t$								
Estimate	0.917	0.041	0.133	0.139	0.727	0.000	0.026	0.974
Std error	(0.200)	(0.109)	(0.076)	(0.125)	(0.108)	(0.000)	(0.034)	(0.034)
Akaike		2.824			2.799			2.820
Bayes		2.887			2.878			2.883
Shibata		2.823			2.798			2.819
Hannan-Quinn		2.850			2.831			2.845
Panel B: $\mu_t = \mu_0 + \mu_1^c \Delta c_t + \sum_{i=1}^8 \mu_i^r r_{PC,i,t}^{ex}$								
Estimate	0.902	0.087	0.000	0.000	0.999	0.000	0.000	1.000
Std error	(0.138)	(0.088)	(0.001)	(0.000)	(0.000)	(0.000)	(0.004)	(0.004)
Akaike		2.842			2.851			2.852
Bayes		2.874			2.899			2.884
Shibata		2.842			2.850			2.852
Hannan-Quinn		2.855			2.870			2.865
Panel C: $\mu_t = \mu_0 + \sum_{i=1}^S \rho_i f_{t+1-i}$								
Estimate	0.891	0.115	0.000	0.000	0.999	0.000	0.016	0.984
Std error	(0.150)	(0.102)	(0.000)	(0.000)	(0.000)	(0.000)	(0.060)	(0.060)
Akaike		2.843			2.850			2.856
Bayes		2.875			2.898			2.888
Shibata		2.843			2.850			2.856
Hannan-Quinn		2.856			2.869			2.869

ARCH(1), GARCH(1,1), and IGARCH(1,1) estimates for consumption growth volatility with different models for the conditional mean. Models estimated via QMLE. Robust standard errors constructed using Newey and West (1987). The table also reports Akaike, Bayesian, Shibata, and Hannan-Quinn information criteria.

quarters.¹⁹ These estimates are almost identical to those uncovered in the previous literature (see, e.g., Table 6 in Chen (2017)). However, the statistically insignificant α (Column (4)) indicates a potential identification failure for the model parameters, including β . This is why we also consider an IGARCH specification that once again yields (Columns (6)–(8)) a statistically significant β and an insignificant α , with point estimates compatible with constant consumption volatility.

However, as shown in both the previous literature (see, e.g., Liu and Matthies (2022) and Bansal, Dittmar, and Kiku (2007)) and in Appendix D, lagged consumption alone does not capture the full extent of consumption predictability. Therefore, in Panel B, we add to the AR(1), as drivers of the conditional mean, the same principal components of asset returns

¹⁹The half-life of a covariance-stationary GARCH (1,1) process is $1 + \log(1/2)/\log(\alpha + \beta)$.

that we found to predict consumption (see Appendix D). The resulting change is striking: When we better control for consumption predictability, the evidence in favor of time-varying volatility vanishes in all the model specifications, including the canonical GARCH(1,1). One cannot reject the hypothesis of constant consumption volatility in any specification.²⁰

Finally, in Panel C, we use the conditional mean of our moving average specification (without the contemporaneous shock) evaluated at its posterior mean. First, as in Panel B, there is no evidence of volatility clustering. Second, the sharply different results in Panel A relative to those in Panels B and C suggest that the AR(1) approximation of the conditional mean is not innocuous for the identification of the volatility process. Moreover, if the AR(1) were the true process, our $MA(\bar{S})$ specification would closely approximate it (as shown in Section II.1) and lead to similar implications for volatility. Empirically, however, using an $MA(\bar{S})$ parametrization for the conditional mean leads to drastically different estimations of the volatility models and shows that the canonical results are driven by the misspecification of the mean.

III.3 Time-series properties of stocks and bonds

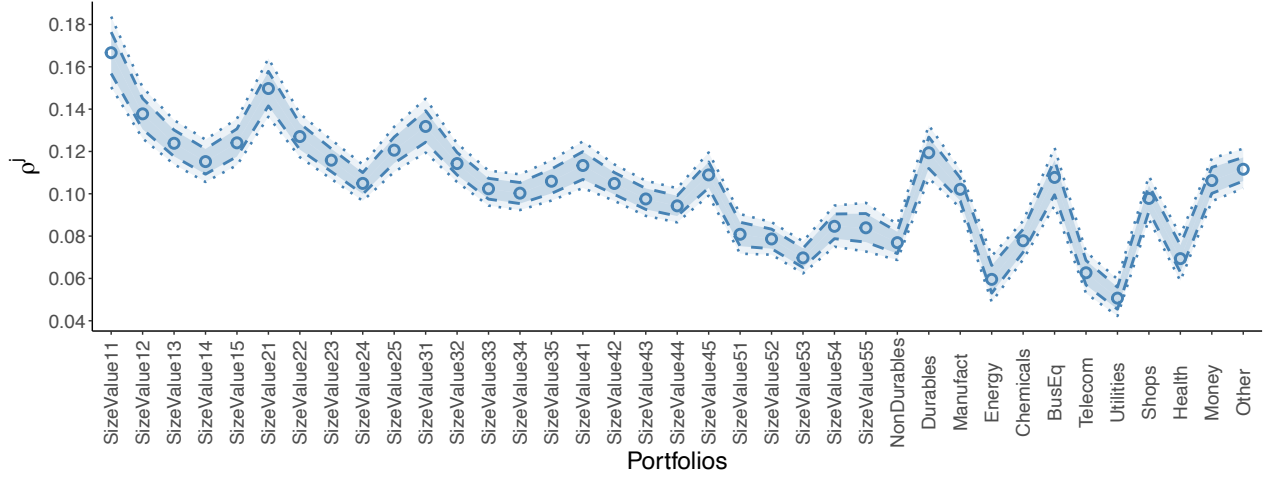
We now turn to the time-series properties of stocks and bonds implied by our model in equations (2)–(3) and (5)–(6). The loadings of equity portfolios on the latent factor f_t are depicted in Figure 10.

The size- and book-to-market-sorted portfolios are ordered first (e.g., Portfolio 2 is the smallest quintile of size and the second smallest decile of book-to-market ratio), followed by the 12 industry portfolios (Portfolios 26–37 in the graph). All the portfolios have significant and positive exposures to the factor driving the conditional mean of consumption growth. Note, however, that this does not imply a single-factor model for returns, as our specification allows for multiple systematic factors in the return space orthogonal to f_t by the identification restriction. The findings remain unchanged when a bond-specific factor is added to the model as in equations (5)–(6) (see Figure IA.23 in the Internet Appendix).

The loadings are not only statistically but also quantitatively significant, as shown in Figure 11: The common (filtered) factor f explains on average 79% of the time-series variation of the 12 industry and 25 Fama and French (1992) size- and book-to-market-sorted

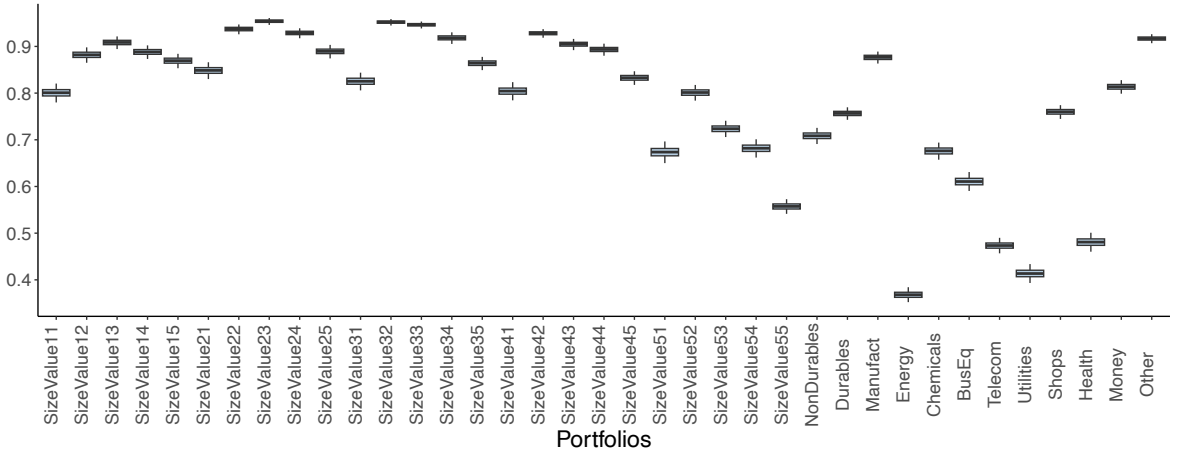
²⁰Note that a (I)GARCH(1,1) with $\alpha = \omega = 0$ and $\beta = 1$ is identical to a constant volatility model.

Figure 10: Common factor loadings (ρ^r) of the stock portfolios in the one-factor model.



Posterior means of the stock factor loadings on f_t (circles) and centered posterior 90% (dashed line) and 68% (dotted line) coverage regions in the one-latent-factor model. Ordering of portfolios: 25 Fama and French (1992) size- and book-to-market-sorted portfolios and 12 industry portfolios.

Figure 11: Share of stock portfolios' return variance explained by the f component.



Box-plots (posterior 95% coverage area) of the percentage of time-series variances of individual stock portfolio returns explained by the f component in the one-factor model. Ordering of portfolios: 25 Fama and French (1992) size- and book-to-market-sorted portfolios and 12 industry portfolios.

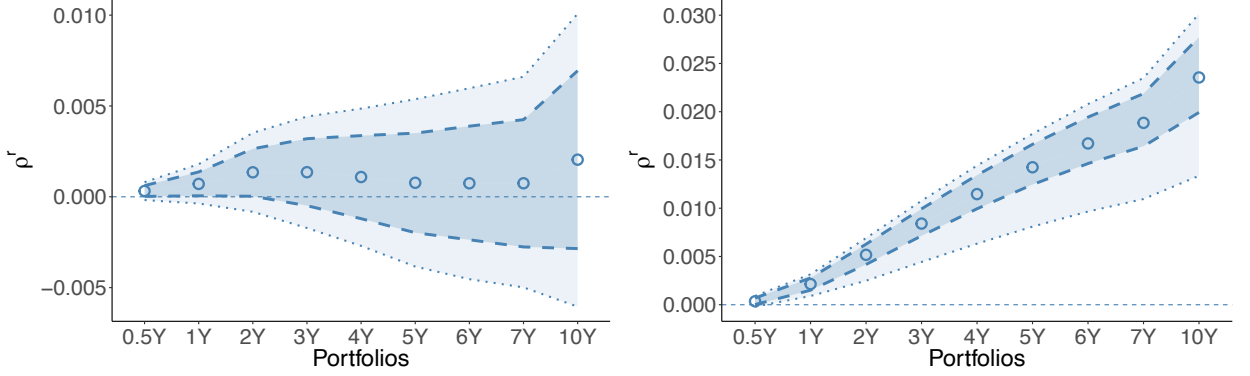
equity portfolios, ranging from 36% to nearly 95% for individual portfolios.²¹

As shown in Figure IA.24 in the Internet Appendix, adding a bond-specific factor leaves the variance decomposition of stock returns basically unaffected.²²

²¹ This observation is robust to a multi-factor model. In Panel A, Figure IA.46 of the Internet Appendix, we report the fraction of return variation explained by the f component when using a five-factor model to extract the conditional mean shocks in nondurable consumption growth. The results therein are extremely similar: the state-space model explains on average 73% of stock return variation.

²² The high explanatory power of the state-space model for asset returns is not a mechanical byproduct of our method. As the simulation studies in Figure 2 suggest, our method can capture the conditional dynamics

Figure 12: Bond loadings on common factor f_t .

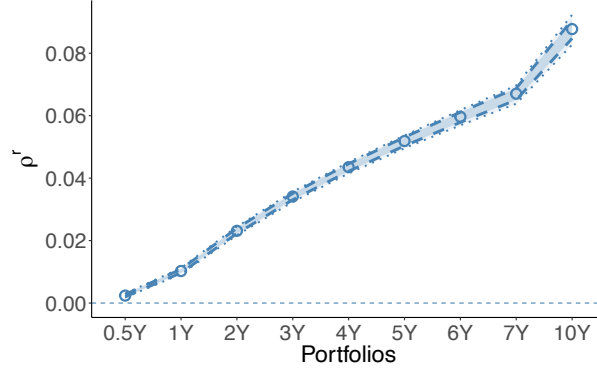


(a) Bond loadings on f_t , one-factor model.

(b) Bond loadings on f_t , two-factor model.

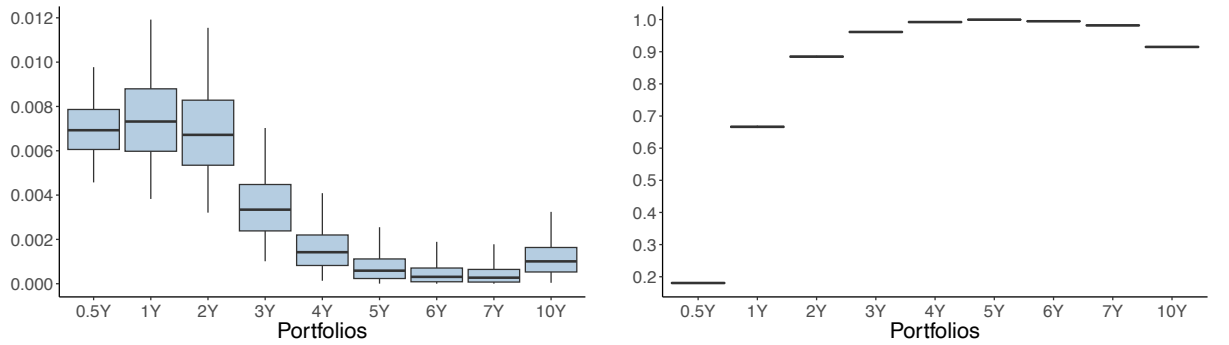
The figure shows posterior means of the bond factor loadings on f_t (blue circles) and centered posterior 90% (dashed line) and 68% (dotted line) coverage regions in the one- and two-factor models.

Figure 13: Bond loadings (θ^b) on the bond-specific factor (b_t).



The figure shows posterior means (blue circles), the centered posterior 90% (dashed line) and 68% (dotted line) coverage regions, of bond loadings on the bond-specific factor b_t .

Figure 14: Variance decomposition box plots of bond returns.



(a) Percentage of the time-series variances of bond returns explained by f_t , one-factor specification.

(b) Percentage of the time-series variances of bond returns jointly explained by f_t and b_t components.

The figure shows box plots (posterior 95% coverage area) of the percentage of time-series variances of bond returns explained by the f_t (left panel) and f_t and b_t (right panel) shocks.

The loadings of the bond portfolios on the common consumption factor f_t are reported in Figures 12a and 12b, respectively, for the one- and two-latent-factor specifications. Both sets of estimates show an upward-sloping term structure of the loadings, and the point estimates are quite similar in the two specifications, with the main difference being that allowing for a bond-specific factor (b_t) delivers much sharper estimates of the loadings on the common factor f_t . The magnitude of these loadings is considerably smaller than that of stocks. Although these numbers may not seem as impressive as those for the cross-section of stocks, the pattern is highly persistent and significant, confirming a common factor structure between nondurable consumption growth and asset returns.

The loadings on the bond-specific factor b_t are reported in Figure 13. These loadings are highly statistically significant and increase steeply and monotonically with maturity, revealing a rather pronounced term structure pattern.

Finally, Figure 14 reports the share of time-series variation of bond returns explained by the f_t shocks (left panel) and the f_t and b_t shocks (right panel). It highlights the importance of allowing for a bond-specific factor to characterize the time-series of bond returns. The common factor f_t accounts for a small (about 1%) but statistically significant proportion of the time-series variation in bond returns. The bond-specific factor, in turn, captures most of the residual time-series variation in returns. Although the model captures almost 55% of the variation in the six-month bond returns, its performance rapidly improves with maturity and results in a nearly perfect fit for time horizon of about five years.

Once again, our setting has an important testable implication that can be verified directly from the data without making all of the ancillary assumptions of the state-space model. In particular, equation (8) implies that, given our state-space results for the consumption loadings in Figure 3, the covariance between asset returns and multi-period consumption growth should display an increase in both its level and cross-sectional dispersion. This conjecture is supported by Figure IA.35 of the Internet Appendix, which depicts $Cov(\Delta c_{t,t+1+S}, r_{j,t+1}^e)$ for various assets j and horizons S . As we move away from the standard case of $S = 0$, two observations immediately arise. First, there is a substantial increase in the average exposure of

even when the conditional mean of consumption growth is determined by small PCs, in which case the f components should explain relatively small shares of return variation. To verify this property of the method in the data, we utilize the same state-space model to extract the f components in the growth rate of real GDP, durable consumption, consumer price index (CPI), and service and report the related results in Panels B, C, D, and E, Figure IA.46 of the Internet Appendix, respectively. On average, the state-space models of GDP, durable consumption, CPI, and service growth explain 69%, 35%, 34%, and 17% of the time-series variation of 37 stock returns.

asset returns to consumption growth. Second, there is a strong “fanning out” effect observed for the higher values of the consumption horizon S . This spread in covariances rationalizes the finding of Parker and Julliard (2005), who use $\Delta c_{t,t+1+S}$ to price time t asset returns.

III.4 External predictors of consumption and returns

The previous analysis assumes that asset returns are serially uncorrelated over time. However, our framework is flexible in incorporating other predictors in consumption growth and asset returns. We use two predictors to introduce serial correlations: the market’s price-dividend ratio and the Chicago Fed National Activity Index (CFNAI). The two-factor model is as follows:

$$\Delta c_{t-1,t} = \mu_c + \sum_{j=0}^{\bar{S}} \rho_j f_{t-j} + \sum_{j=0}^{\bar{S}} \theta_j b_{t-j} + \gamma_1^c p d_{t-1} + \gamma_2^c c f n a i_{t-1} + w_t^c, \quad (13)$$

$$\text{Stocks: } r_{st}^e = \mu_{sr} + \rho^{sr} f_t + \gamma^s p d_{t-1} + w_t^{sr}, \text{ and} \quad (14)$$

$$\text{Bonds: } r_{bt}^e = \mu_{br} + \rho^{br} f_t + \theta^b b_t + \gamma^b c f n a i_{t-1} + w_t^{br}, \quad (15)$$

where the $p d_{t-1}$ is the lagged price-dividend ratio of the market portfolio and $c f n a i_{t-1}$ is the lagged CFNAI.²³ In addition, we allow these two predictors to drive the conditional dynamic of consumption growth, with the goal of testing whether they can crowd out the importance of the MA component of f_t .

We first test whether the P/D ratio and CFNAI can predict consumption growth and asset returns. Note that the CFNAI has been available since January 1967, so the sample used in this subsection differs slightly from that of other parts. Figure IA.25 in the Internet Appendix plots the distribution of γ^c , γ^s , and γ^b in equations (13)–(15). While the P/D ratio predicts neither consumption growth nor stock returns, CFNAI is essential in characterizing the conditional dynamics of longer-maturity bonds (3Y–10Y) and consumption growth. Because we standardize the predictors to have unit variances, the estimate implies that if CFNAI rises by one standard deviation, consumption growth will increase by 0.13% quarterly, with a 95% confidence interval of [0.01%, 0.25%].

Next, we study the identification of f_t shock and consumption’s impulse responses. The

²³CFNAI is a weighted average of 85 monthly indicators of national economic activity, delivering a rough measure of a common factor in these national economic data. In fact, this measure is similar to the canonical measure proposed by James Stock and Mark Watson, which has been used to predict inflation since the late-1990s.

empirical results, as shown in Figure IA.27, are reassuring: We still detect the slow adjustment of consumption growth to f_t shock, although the size slightly declines as CFNAI captures part of the predictability. Moreover, the MA component, loadings on f_t , and variance decompositions of consumption growth have almost identical patterns as previously.

III.5 Multifactor analysis

The simulation study in Section II.2 shows that our framework can identify the conditional dynamic of consumption growth even if the common component, f_t , is a small PC of asset returns or a linear combination of several PCs. To ensure that the single-factor analysis in the previous sections does not neglect important information in returns, we now consider a five-factor model for asset returns in equations (2)–(3).

Figure IA.29 in the Internet Appendix plots the related results. First, all the important estimates, including the MA component of consumption growth, impulse responses to f_t shock, stocks and bonds' loadings on f_t shock, and the variance decomposition of predicting consumption growth, are almost unchanged using the five-factor model. Hence, the single-factor model that we considered before can characterize the key patterns of the consumption mean process in the simplest form. Second, it is worth noting that the multifactor model introduces additional estimation noise. For instance, predictive R_{adj}^2 in Panels (e) and (f) are centered around the same estimates yet have wider posterior coverage than those in the single-factor case. As more latent factors do not identify any additional useful information but do increase estimation errors, we focus on the more parsimonious single-factor model in the following section and report the multifactor results as robustness checks.

III.6 Short- and long-run stochastic volatilities

Although our state-space model for consumption and returns in equations (2)–(3) delivers consistent estimates of the conditional mean parameters even in the presence of time-varying volatility for the error terms, the literature has often focused on models with a joint SV process for consumption and asset returns. This approach has been commonly used because it provides time-varying risk premia for equilibrium asset returns in representative agent models.

As shown in Section III.2, ad hoc specifications for the mean process (e.g., constant or AR(1)), which miss the true degree of predictability in the consumption process, mechanically

invalidate the identification of the consumption volatility process and its predictability. This is why we rely on a long MA representation that can correctly capture the conditional mean dynamics irrespective of its true functional form and, therefore, provides a robust way to assess the nature and properties of the consumption volatility without taking a stand on the data-generating process of the conditional mean. That is, our state-space representation provides a reliable framework to manage inference on a class of structural models rather than a particular parametrization.

We generalize our state-space model in equations (2)–(3) to allow for stochastic volatilities in *all* innovations. Furthermore, we allow the volatilities to drive time-varying risk premia; that is, the process for log excess returns is now

$$\underset{N \times 1}{\mathbf{r}_t^e} = \underset{N \times 1}{\boldsymbol{\mu}_r} + \underset{N \times 1}{\boldsymbol{\rho}^r} f_t + \underset{N \times 1}{\boldsymbol{\beta}_c} \sigma_{c,t-1}^2 + \underset{N \times 1}{\boldsymbol{\beta}_f} \sigma_{f,t-1}^2 + \underset{N \times 1}{\boldsymbol{\beta}_r} \sigma_{r,t-1}^2 + \underset{N \times 1}{\mathbf{w}_t^r}, \quad (16)$$

where $\sigma_{c,t-1}$ is the stochastic volatility of w_t^c ; $\sigma_{f,t-1}$ is the stochastic volatility of f_t , the innovation to expected consumption growth; and $\sigma_{r,t-1}$ is a common market volatility process that affects the volatility of all excess return shocks ($\mathbf{w}^r$). In the language of Schorfheide, Song, and Yaron (2018), $\sigma_{c,t-1}$ and $\sigma_{f,t-1}$ are, respectively, the so-called short- and long-run volatilities of consumption, and the latter captures the time-varying volatility of expected consumption growth.

We follow the past literature (e.g., Hull and White (1987) and Chesney and Scott (1989)) by assuming the following distributions for w_t^c , f_t , and $\mathbf{w}_t^r$:

$$w_t^c \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \underbrace{\exp(h_{ct})}_{\sigma_{c,t-1}^2}\right), \quad f_t \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \underbrace{\exp(h_{ft})}_{\sigma_{f,t-1}^2}\right), \quad \text{and} \quad (17)$$

$$\mathbf{w}_t^r \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{r,t-1}), \quad \boldsymbol{\Sigma}_{r,t-1} = \text{diag}\{\sigma_{r,1,t-1}^2, \dots, \sigma_{r,N,t-1}^2\}, \quad (18)$$

where $\sigma_{r,i,t-1}^2$ is the stochastic volatility of the i -th asset that is driven by both idiosyncratic volatility shocks ($e_{i,rt}$) and the common volatility process ($\sigma_{r,t-1}^2 \equiv \exp(h_{rt})$); that is,

$$\sigma_{r,i,t-1}^2 = \exp\{\kappa_0^{(i)} + \kappa_1^{(i)} h_{rt} + e_{r,i,t}\}. \quad (19)$$

In what follows, for simplicity, we refer to $\sigma_{r,t-1}^2 \equiv \exp(h_{rt})$ as the “market” variance, albeit in our formulation of asset return dynamics, the total volatility of market returns is a combination of $\sigma_{r,t}$ and $\sigma_{f,t}$.

The formulation above is rather general and encompasses the standard models in the literature as particular cases. For instance, in the canonical long-run risk model of Bansal

and Yaron (2004), the short- and long-run volatilities of consumption are identical ($\sigma_{c,t}^2$ is proportional to $\sigma_{f,t}^2$), with the consumption dynamics summarized as follows:

$$\Delta c_{t-1,t} = \mu_c + x_{t-1} + w_t^c, \quad w_t^c \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{c,t-1}^2), \quad \text{and} \quad x_t = \rho_x x_{t-1} + \varphi_e \sigma_{t-1} e_t, \quad e_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

which implies that $x_t = \varphi_e \sum_{j=0}^{\infty} \rho_x^j \sigma_{t-j-1} e_{t-j}$, where σ_t is the only time-varying economic uncertainty. This formulation can be mapped into our general framework for the consumption dynamics in equation (2), imposing the restrictions i) $\rho_0 = 0$ and $\rho_j = \varphi_e \rho_x^{j-1}$ and ii) $f_t = \sigma_{t-1} e_t$. Hence, in this case, w_t^c and f_t follow the same SV process.

Schorfheide, Song, and Yaron (2018) further separate the volatility of consumption into short- and long-run components, as follows:

$$\Delta c_{t-1,t} = \mu_c + x_{t-1} + \sigma_{c,t-1} \eta_{c,t} \quad \text{and} \quad x_t = \rho_x x_{t-1} + \sqrt{1 - \rho_x^2} \sigma_{x,t-1} \eta_{x,t},$$

where $\sigma_{i,t}$ ($i \in \{c, x\}$) denotes two distinct stochastic volatility processes, and $\eta_{c,t}$ and $\eta_{x,t}$ follow uncorrelated standard normal distributions. In this formulation, $\sigma_{c,t}$ ($\sigma_{f,t}$) is meant to capture the short-run (long-run) volatility in consumption. This setup also fits within our general MA formulation in equation (2) after imposing the following restrictions: i) $w_t^c = \sigma_{c,t-1} \eta_{c,t} \sim \mathcal{N}(0, \sigma_{c,t-1}^2)$, ii) $x_t = \sqrt{1 - \rho_x^2} \sum_{j=0}^{\infty} \rho_x^j \sigma_{x,t-j-1} \eta_{x,t-j}$, iii) $\rho_0 = 0$ and $\rho_j = \sqrt{1 - \rho_x^2} \rho_x^{j-1}$, and iv) $f_t = \sigma_{x,t-1} \eta_{x,t}$ and $\sigma_{f,t} = \sigma_{x,t}$.

In order to identify the model, without loss of generality, we normalize h_{ft} and the common (log) market volatility component h_{rt} to have zero mean and unit variance. The stochastic volatility processes h_{ct} , h_{ft} , and h_{rt} are all modeled as independent AR(1) processes in our unrestricted formulation. See Internet Appendix B.2 for further details, including the MCMC algorithm needed to evaluate this model.

Using our general stochastic volatility formulation, the estimated parameters for the consumption ($\{\rho_j\}_{j=0}^S$) and asset returns ($\boldsymbol{\rho}^r$) loadings on the common shock f remain essentially unchanged (see Figure IA.30 in the Internet Appendix). Similarly, the variance decompositions for consumption and returns are almost identical to those presented in the previous sections. Although this is not surprising (because the formulations (2)–(3) are consistent for mean equation parameters independently from the volatility process), this finding is quite reassuring. In other words, it is clear that the returns significantly load on the shocks to the consumption mean. These shocks drive more than a quarter of the consumption growth variance. Overall, all of our results are robust to the modeling choice for volatility.

What do we learn about the volatility itself for both consumption and returns? Figure

Table 4: Correlations among stochastic volatility processes and VXO index

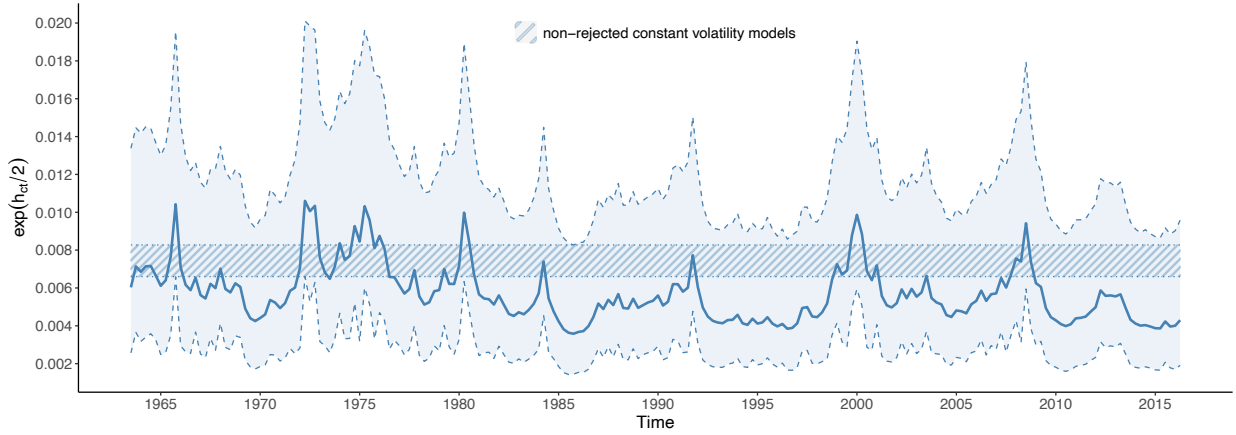
	Mean	2.5%	5%	50%	95%	97.5%
Panel A: correlations of vol processes with VXO ² index						
$cor(\sigma_{ct}^2, VXO_t^2)$	0.18	-0.03	-0.00	0.18	0.39	0.43
$cor(\sigma_{ft}^2, VXO_t^2)$	0.45	0.22	0.25	0.45	0.62	0.65
$cor(\sigma_{rt}^2, VXO_t^2)$	0.52	0.37	0.40	0.53	0.62	0.63
Panel B: Pairwise correlations of vol processes						
$cor(\sigma_{ct}^2, \sigma_{ft}^2)$	0.07	-0.09	-0.07	0.06	0.26	0.30
$cor(\sigma_{ct}^2, \sigma_{rt}^2)$	0.15	-0.02	0.00	0.14	0.32	0.36
$cor(\sigma_{ft}^2, \sigma_{rt}^2)$	0.22	0.06	0.08	0.21	0.38	0.41

The table summarizes posterior mean, 2.5%, 5%, 50%, 95%, and 97.5% quantiles of correlation among σ_{ct}^2 , σ_{ft}^2 , σ_{rt}^2 , and the VXO index under a diffuse prior for the autoregressive coefficients of the vol processes.

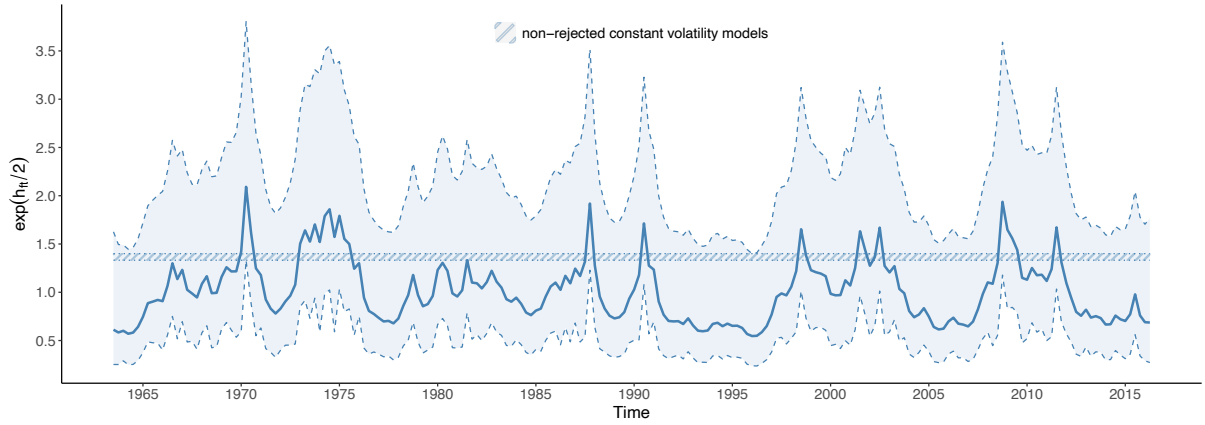
15 reports the estimated volatility processes (posterior median and 95% credible intervals) under a diffuse prior for the autoregressive coefficients of the processes. Several observations are in order. First, there is clear evidence of time-varying volatility in stock returns (Panel C), whereas our findings are more nuanced for the consumption volatilities. Second, for the short-run consumption volatility ($\sigma_{c,t}$ in Panel A), there is an entire range of constant values that are within the posterior confidence bands. This finding holds even when we consider monthly consumption data as well as the mixed-frequency state-space formulation that accounts for time-aggregation bias, as we show in the Internet Appendix (see, respectively, Figures IA.6 and IA.12). Third, instead, while the estimated long-run consumption volatility ($\sigma_{f,t}$) in Panel B of Figure 15 contains an entire range of constant values within the posterior credible intervals, there is no such parameter region i) using monthly consumption data (see Figure IA.6), ii) estimating a mixed-frequency model (see Figure IA.12), and iii) directly incorporating a leverage effect into $\sigma_{f,t}$ (see Figure IA.16).²⁴ In other words, the long-run consumption volatility becomes more sharply identified in latter three cases. Note also that the confidence bands in both Panels A and B are sufficiently wide to allow for potential substantial time variation in both short-run and long-run consumption volatilities. Fourth, all asset volatilities have significant loadings (κ_1^i) on the common financial market volatility process (see Figure IA.45 of the Internet Appendix). Fifth, the common financial market volatility increases during recessions and market crashes, whereas short- and long-run consumption volatilities do not display such a clear pattern.

²⁴ We extend our framework to allow for a leverage effect in the long-run consumption volatility following Omori, Chib, Shephard, and Nakajima (2007). Details are reported in Internet Appendix J.

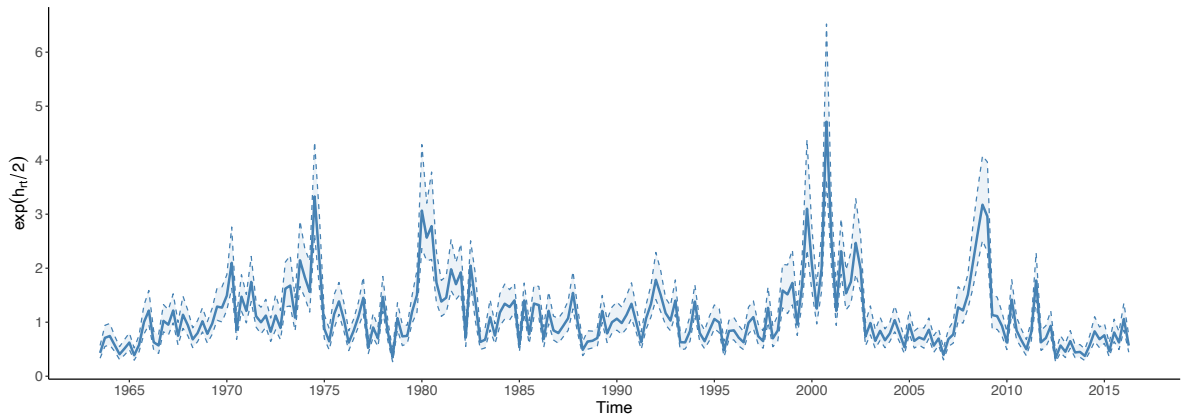
Figure 15: Filtered stochastic volatilities of consumption and returns.



Panel A: Log volatility of the short-run consumption shock (w_t^c) of equation (2)



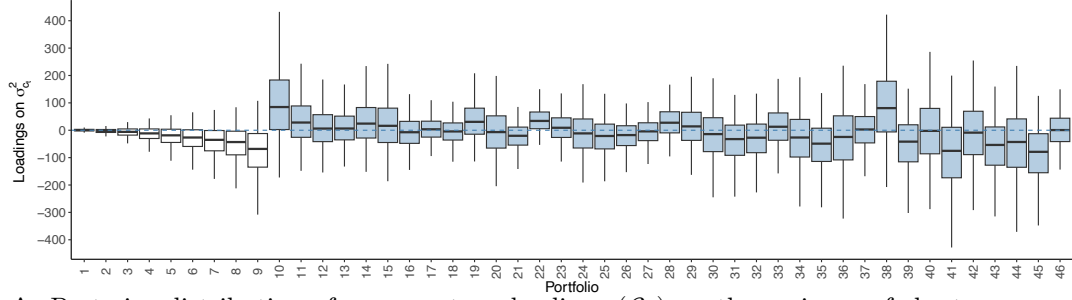
Panel B: Log volatility of the shock to the conditional mean of consumption growth (f_t).



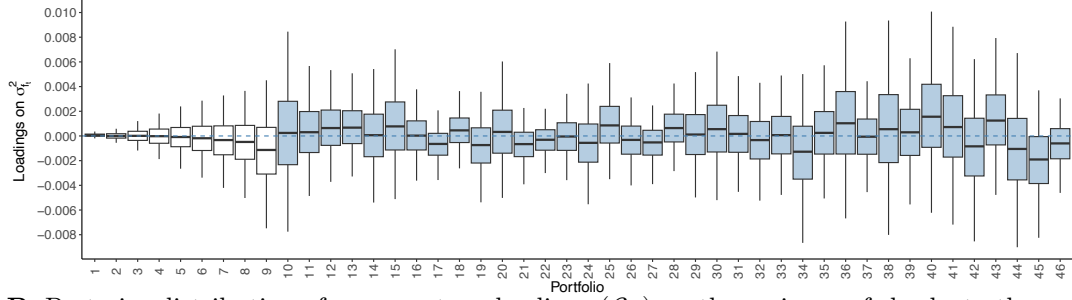
Panel C: Common log volatility of asset return (h_{rt} of equation (19)).

Estimated stochastic volatilities of the model in equations (2), and (16)–(19), and Section B.2 under a diffuse prior for the autoregressive volatility coefficients. Solid blue lines depict the posterior median of the log volatility, whereas dotted red lines denote 2.5% and 97.5% credible intervals. Shaded (patterned) areas reflect constant volatility levels that would not be rejected given the credible intervals.

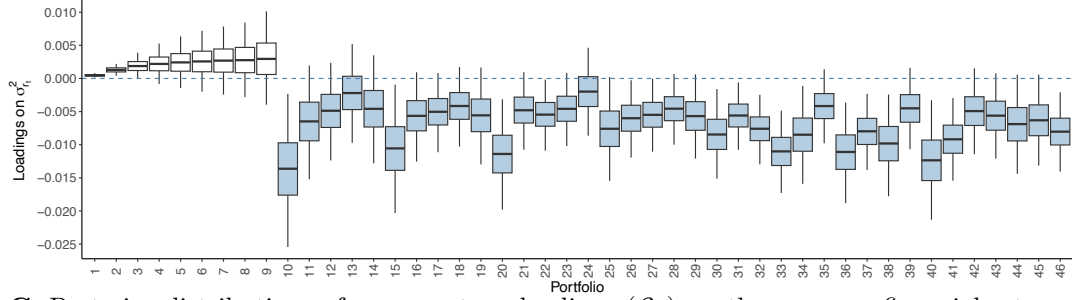
Figure 16: Loadings of excess returns on consumption and returns volatilities.



Panel A: Posterior distribution of excess return loadings (β_c) on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$) in equation (16).



Panel B: Posterior distribution of excess return loadings (β_f) on the variance of shocks to the conditional consumption growth mean ($\sigma_{f,t-1}^2$) in equation (16).



Panel C: Posterior distributions of excess return loadings (β_r) on the common financial return variance ($\sigma_{r,t-1}^2$) in equation (16).

This figure shows the box plots of the posterior distributions of the loadings of portfolio excess returns on the variance of short-run consumption shocks ($\sigma_{c,t-1}^2$), the variance of shocks to the conditional consumption growth ($\sigma_{f,t-1}^2$), and the common financial returns variance ($\sigma_{r,t-1}^2$). Portfolios are ordered with bonds first (1–9), Fama-French 25 size and book-to-market second (10–34), and industry portfolios last.

We further explore the correlation between the estimated volatilities and (the square of) the VXO volatility index (a proxy for the underlying volatility process commonly used in the literature) in Table 4.²⁵ Both the financial market and long-run consumption volatilities

²⁵We use the VXO index, rather than VIX, due to the longer time-series available for the former. The VXO (as is the VIX), measured in discrete time, is adapted to both the physical underlying volatility process and the “true” latent stochastic discount factor. Hence, the correlations we measure are potentially generated by either or both of them.

have nontrivial correlations with VXO (.53 for the former and .45 for the latter, at the median, in Panel A), while being weakly correlated with each other (the median correlation is only .21, Panel B). These correlations give an estimate of the variance decomposition for the S&P100 (the index underling the VXO). Hence, Panel A of Table 4 implies that about 45% of the (option-implied) variance of the S&P100 index is generated by f_t , the shocks to the conditional mean of consumption. This is a strong external verification of our (rather general) modeling choice for the joint dynamics of consumption and returns.

We provide additional external validation of our estimated volatility processes in Table IA.XII of the Internet Appendix, which reports the correlation between the filtered stochastic volatilities and the real/macro/financial uncertainty measures in Jurado, Ludvigson, and Ng (2015) and Ludvigson, Ma, and Ng (2021). We find that the long-run consumption and asset-market volatilities significantly load on all three uncertainty indices. On the contrary, the short-run consumption volatility weakly correlates with these uncertainty measures at best.

Overall, our findings suggest, as in Schorfheide, Song, and Yaron (2018), different dynamics for the short- and long-run consumption volatilities, but also a distinct process for the market common volatility.

But do the volatilities of returns and consumption drive excess return dynamics? Figure 16 reports the posterior distributions of the loadings in equation (16). Strikingly, none of the excess return series loads significantly on short- and long-run consumption volatilities (Panels A and B). This finding is robust even when we i) use monthly consumption data, ii) estimate a mixed-frequency model, and iii) directly incorporate a leverage effect into $\sigma_{f,t}$ (see, respectively, Figures IA.7, IA.13 and IA.17 of the Internet Appendix). This evidence poses a challenge to the literature that has modeled time-varying risk premia by assuming time variation in the volatility process of consumption. Panel C shows some support for the common financial market volatility being negatively associated with excess returns. Nevertheless, this latter finding is generally weak and fragile with respect to the various generalizations that we consider (see Figures IA.7, IA.13, and IA.17 of the Internet Appendix).

The generality of our setting also allows us to formally evaluate a multiplicity of alternative formulations considered in the previous literature. We do so in Section I of the Internet Appendix. The data favor a specification in which i) excess returns do not load on *any* of the stochastic volatilities and ii) short- and long-run consumption volatilities are different

from each other and distinct from the common market volatility – the posterior probability of such a formulation is almost 100%.

The finding that excess returns do not significantly load on the volatilities is perhaps not too surprising, given the mixed and inconclusive evidence on the volatility-return trade-off in the literature (see, e.g., French, Schwert, and Stambaugh (1987), Nelson (1991), and Campbell and Hentschel (1992)). However, Bandi and Perron (2008) show that this tradeoff becomes stronger at longer horizons. Motivated by this, we explore whether the connection between returns and volatilities strengthens at lower frequencies. In particular, we examine whether the common market and (short- and long-run) consumption volatilities predict multi-horizon excess returns.²⁶ Predictive regression coefficients for four- to 12-quarter cumulative returns are reported in Figures IA.42–IA.44 of the Internet Appendix. Overwhelmingly, we find insignificant coefficients at all horizons. That is, volatilities do not seem to predict excess returns even at these lower frequencies.

III.7 Further Robustness Checks

We conduct an extensive set of additional robustness checks and show that all our key results presented in the previous sections remain unchanged. In this section, we discuss the main exercises and report additional ones in Section K of the Internet Appendix.

First, one may be concerned that our empirical results strongly depend on a particular choice of the MA length of the consumption mean process, $\bar{S}$. The data indicate that this is not the case. When the MA component includes at least six quarters of data, our results remain almost unchanged. Figure A1 in the Appendix reproduces all the main findings for the state-space formulation estimated with $\bar{S} = 20$ quarters, that is, five years of data, and shows that they are qualitatively and quantitatively identical to those presented in the previous sections. We find that the similar size of the cumulative impulse response of consumption is approximately 1%. The reason for this similarity is that the incremental consumption impulse response to f_t is almost zero after two to three years. Note that this finding is consistent with the evidence in Bandi and Tamoni (2023), who show that most of the (priced) consumption risks occur within the business cycle frequency.

²⁶Given the AR(1) nature of our log stochastic volatility processes, this exercise is equivalent to investigating whether the conditional variance of the multi-period consumption, $\text{var}_t(\Delta c_{t,t+H})$ ($H > 1$), can forecast cumulative excess returns, $\mathbf{r}_{t,t+H}^e$.

Second, because the state-space estimation results may depend on the choice of the cross-section of assets used for the analysis, we repeat our analysis using a multiplicity of alternative base assets, as discussed in Section K of the Internet Appendix. In particular, Figures IA.34, IA.40, and IA.41 of the Internet Appendix present all the key empirical findings of the paper, estimated on a wide cross-section of characteristic-based anomaly portfolios used in Kozak, Nagel, and Santosh (2020) (KNS). Our findings either remain virtually unchanged, or become even stronger, which again emphasizes their robustness to the choice of test assets.

To summarize, our findings are robust to a plethora of perturbations to the baseline specification, all supporting the same key takeaways: *i*) the conditional mean of consumption growth is spanned by returns; *ii*) consumption reacts to financial market innovations slowly, with the total cumulative effect peaking after two to three years; *iii*) short- and long-run consumption shocks and returns have distinct volatility processes, and neither seems to drive time-varying risk premia.

IV Asset Pricing Implications

The first-order importance of financial markets in driving the consumption process that we have uncovered raises the natural question of whether the conditional mean shocks command a significant risk premium. Furthermore, can our identified consumption process improve the performance of basic consumption-based representative-agent models with lower risk aversion? Can we also explain the level and volatility of the risk-free rate? That is, can we jointly provide an explanation of both the equity premium (Mehra and Prescott (1985)) and risk-free rate (Weil (1989)) puzzles? The answers to all of these questions, as illustrated in the next two subsections, are affirmative.

IV.1 The risk premium of consumption mean shocks

We now turn to assessing the cross-sectional pricing of the shocks to the consumption mean. We do not take a stand on the preferences determining risk prices, as we only want to assess whether the f_t shocks are part of the stochastic discount factor (SDF). Given the evidence in Bryzgalova, Huang, and Julliard (2023) that the SDF in the economy is *dense* in the space of observable factors, we need to account for the omitted variables. We do so following Giglio and Xiu (2021) and Bryzgalova, Huang, and Julliard (2024); that is, we maintain the

assumption about asset return dynamics in equation (3) and further postulate the following log linear SDF:

$$\Delta \mathbb{E}_t[m_t] := m_t - \mathbb{E}_{t-1}[m_t] = - \underbrace{b_f f_t}_{CCAPM} - \underbrace{\mathbf{b}_g^\top \mathbf{g}_t}_{omitted\ factors}, \quad (20)$$

where b_f is the risk price of the shock to conditional consumption mean, and the second component, $\mathbf{b}_g^\top \mathbf{g}_t$, is included to account for the omitted factor bias in estimating f_t 's risk price.²⁷ Also note that only the shocks to the SDF, $\Delta \mathbb{E}_t[m_t]$, matter for the pricing excess returns. Furthermore, since f_t is normalized to have unit variance and is (by construction) orthogonal to the omitted factors, b_f is both its risk premium, price of risk, and (tradable) Sharpe ratio.

Table 5 reports the estimates of f_t 's risk premium in several cross-sections of excess returns. First, f_t , the shock to both asset returns and conditional consumption mean, carries a stable and significant risk premium in all cross-sections. Hence, portfolios with higher consumption risk exposure, on average, earn higher returns. Furthermore, our estimates of the quarterly risk premium (b_f) are virtually identical to the ones delivered by the misspecification-robust estimator of Giglio and Xiu (2021). Note that both Bayesian and frequentist approaches recover the risk premium as defined by the negative covariance of the factor(s) with the tradable SDF (spanned by latent tradable factors). By doing so, both estimators are robust to weak identification and model misspecification.

Second, the Sharpe ratio of the consumption risk is about 0.44-0.55 per annum. Nevertheless, there are additional sources of risk driving the cross-section of asset returns: the Sharpe ratio of the entire SDF (which further includes the latent factors $\mathbf{g}_t$) is about 0.8–1.0. The column reporting $\mathbb{E} \left[\frac{SR_f^2}{SR_m^2} \mid \text{data} \right]$ – the share of SDF variance explained by consumption mean shocks – further quantifies the importance of consumption risk in the SDF. Across panels, consumption risk explains 22-37% of the total SDF variance. In other words, consumption risk is a first-order component of the SDF, but there is also a significant amount of additional shocks, orthogonal to consumption risk, that drive the cross-section of asset returns.

Third, our SDF has high explanatory power for all the cross-sections, as indicated by both the high cross-sectional R^2 in the 41-86% range, and extremely small and insignificant average pricing errors.

²⁷We can interpret $\mathbf{b}_g^\top \mathbf{g}_t$ as the missing component in C-CAPM as in Ghosh, Julliard, and Taylor (2016).

Fourth, the estimated SDF can almost perfectly capture the 7% average annual market risk premium²⁸ (column $-\text{cov}(r_t^{mkt}, m_t)$). And crucially, it does so almost entirely thanks to the consumption risk component (column $-\text{cov}(r_t^{mkt}, -b_f f_t)$).

Finally, the results in Table 5 appear to be extremely robust. First, a comparison of the results in Panels A and D highlights that the estimates of consumption risk premia are not driven by the differential exposures of stocks and bonds to the f_t shocks— removing bonds does not affect risk premia estimates (see Panel D). Second, using a wide cross-section of stock anomalies (the ones of Kozak, Nagel, and Santosh (2020) in Panel B) yields almost identical risk premia estimates as with our baseline cross-section (in Panel A). Third, a comparison of Panel B vs. Panel C, and of Panel D vs. E, highlights that both the Bayesian (Bryzgalova, Huang, and Julliard (2024)) and frequentist (Giglio and Xiu (2021)) robust estimation methods are not affected by span-invariant repackaging of the test assets. Internet Appendix N presents estimates based on the frequentist Fama-MacBeth regressions. In contrast to the robust estimators of Table 5, risk premia estimates produced by Fama-MacBeth regressions are very sensitive (as theoretically expected) to model parametrization and repackaging of the test assets. This, however, is not a property of the shocks to consumption mean and the risk premia they carry, but an artifact of the fragility of Fama-MacBeth estimation (see the discussion in Internet Appendix N).

But what type of returns predict consumption? That is, what type of portfolios and which industries are strongly correlated with the conditional consumption mean shocks?

Table 6 reports the correlation of the filtered f_t shocks with common tradable risk factors and leading principal components of returns for two different cross-sections of base assets: our baseline one and the KNS anomaly portfolios. First and foremost, we observe a strong and stable correlation of the consumption innovations with the overall market index (81 – 92%) and the first principal component of the based assets (85 – 96%). That is, financial market-wide shocks are the main drivers of the consumption mean process. This explains why, in Table 5, the conditional mean shocks can, at least in reduced form, generate a risk premium that almost perfectly matches the historical mean excess returns on the market index. Second, the returns on small firms are particularly salient, with large loadings on f_t (see Figure 10) declining monotonically with the size characteristic of the portfolios. This, in turn, leads to a strong correlation in Table 6 with the SMB factor (about 60%

²⁸In the sample between 1963:Q3 and 2019:Q4, the time-series average of log market excess returns, $\mathbb{E}[r_t^m] + 0.5\text{var}(r_t^m)$, is around 6.8% per year.

Table 5: Cross-sectional estimates of risk premia of conditional consumption mean shocks

Estimating $\Delta \mathbb{E}_t[m_t] = -b_f f_t - b_d^* g_t$							$\mathbb{E}[r_t^{mkt}] =$		GX (2021)
Posterior	Average α	b_f	$\mathbb{E}[SR_m \mid \text{data}]$	$\mathbb{E}[SR_f \mid \text{data}]$	$\mathbb{E}\left[\frac{SR_f^2}{SR_m^2} \mid \text{data}\right]$	R^2	$-\text{cov}(r_t^{mkt}, m_t)$	$-\text{cov}(r_t^{mkt}, -b_f f_t)$	
Panel A. 37 value-size-industry stock and nine bond portfolios in a five-factor model									
Median	0.000	0.236	0.806	0.474	0.371	0.772	0.069	0.070	0.245
90% CIs	[0.000, 0.000]	[0.079, 0.385]	[0.554, 1.128]	[0.180, 0.771]	[0.056, 0.756]	[0.477, 0.903]	[0.033, 0.109]	[0.025, 0.114]	(3.698)
Panel B. 74 Kozak, Nagel, and Santosh (2020) decile 1 and decile 10 anomaly portfolios in a five-factor model									
Median	0.000	0.244	0.960	0.489	0.263	0.409	0.071	0.069	0.221
90% CIs	[0.000, 0.000]	[0.109, 0.382]	[0.735, 1.200]	[0.220, 0.763]	[0.059, 0.563]	[0.213, 0.580]	[0.034, 0.111]	[0.030, 0.110]	(3.375)
Panel C. 37 Kozak, Nagel, and Santosh (2020) decile 10 portfolios plus 37 long-short portfolios in a five-factor model									
Median	-0.002	0.219	0.936	0.437	0.223	0.795	0.087	0.065	0.234
90% CIs	[-0.003, -0.002]	[0.069, 0.358]	[0.710, 1.170]	[0.139, 0.716]	[0.026, 0.550]	[0.657, 0.875]	[0.050, 0.125]	[0.020, 0.109]	(3.536)
Panel D. 37 value-size-industry stock portfolios in a five-factor model									
Median	0.000	0.219	0.715	0.438	0.399	0.449	0.068	0.063	0.210
90% CIs	[0.000, 0.000]	[0.096, 0.344]	[0.493, 0.938]	[0.192, 0.689]	[0.087, 0.770]	[0.050, 0.718]	[0.030, 0.106]	[0.027, 0.100]	(3.268)
Panel E. Transformed 37 value-size-industry stock portfolios in a five-factor model									
Median	0.001	0.276	0.944	0.553	0.354	0.863	0.079	0.078	0.225
90% CIs	[0.000, 0.001]	[0.134, 0.417]	[0.715, 1.170]	[0.269, 0.833]	[0.096, 0.670]	[0.666, 0.941]	[0.040, 0.116]	[0.038, 0.118]	(3.518)

Estimation results for five cross-sections of excess returns: 37 value-size-industry stock and nine bond portfolios (Panel A), Kozak, Nagel, and Santosh (2020) 74 decile 1 and 10 characteristic-sorted portfolios (Panel B), Kozak, Nagel, and Santosh (2020) 37 decile 10 portfolios plus 37 long-short (decile 1 minus decile 10) portfolios (Panel C), 37 value-size-industry stock portfolios (Panel D), and transformed 37 value-size-industry stock portfolios (Panel E). In the transformed cross-section of 37 value-size-industry stock portfolios (Panel E), we long portfolios 1–19; for the remaining 18 portfolios, we construct the long-short portfolios by buying portfolios 20–37 but short-selling portfolios 1–18. We report: (1) average cross-sectional mispricing (α); (2) risk price of the shock to the conditional consumption mean f_t (b_f); (3) annualized Sharpe ratio of the SDF in equation (20), defined as the annualized volatility of the SDF (SR_m); (4) annualized Sharpe ratio of $b_f f_t$ (SR_f); (5) ratio of SR_f^2 to SR_m^2 ; (6) cross-sectional R^2 ; (6) (annualized) market risk premium implied by the SDF, $-\text{cov}(r_t^{mkt}, m_t)$; (7) (annualized) market risk premium implied by the covariance between market excess return and $-b_f f_t$, $-\text{cov}(r_t^{mkt}, -b_f f_t)$; (8) risk premia estimates and their t -statistics based on the three-pass procedure in Giglio and Xiu (2021), with corresponding standard errors. We estimate the risk prices using the Bayesian approach of Bryzgalova, Huang, and Julliard (2024). Details are provided in Internet Appendix M.

in both cross-sections). Third, the correlation with the value anomaly (HML) is small and not statistically significant due to the U-shaped pattern of loadings in Figure 10 in the Growth/Value characteristic space. Fourth, the correlations with both the Robust-Minus-Weak (operating profitability) portfolio and also the Conservative-Minus-Aggressive (investment) portfolio are significantly negative and sizable (respectively, 22 – 30% and 20 – 30%). Fifth, there is more than solely the first PC of returns reflected in the consumption process, with higher-order PCs being often significant predictors of consumption growth (PC4 in Panel A and both PC2 and PC5 in Panel B).

Figure 10 also reports the loadings of the 12 industry portfolios on the consumption mean shocks. All industries have statistically significant and strong-to-moderate loadings on the conditional consumption mean innovations. In particular, Consumer Durables, Business Equipment, and Finance (Money) display the highest coefficients (11 – 12%) while Utilities, Energy, and Telephones and Television Transmission (Telecom) have the lowest (5 – 6%).

Table 6: What drives the consumption shocks spanned by financial markets?

Panel A: Baseline stock and bonds cross-section					
	MKT	SMB	HML	RMW	CMA
Correlation	0.917	0.605	-0.127	-0.221	-0.295
95% CI	[0.863, 0.959]	[0.515, 0.700]	[-0.292, 0.087]	[-0.267, -0.173]	[-0.425, -0.126]
	PC1	PC2	PC3	PC4	PC5
Correlation	0.964	0.072	-0.007	-0.132	-0.024
95% CI	[0.922, 0.990]	[-0.079, 0.258]	[-0.136, 0.120]	[-0.259, -0.048]	[-0.048, 0.000]
Panel B: Kozak, Nagel, and Santosh (2020) 74 anomaly portfolios					
	MKT	SMB	HML	RMW	CMA
Correlation	0.811	0.601	-0.0579	-0.299	-0.200
95% CI	[0.71 , 0.909]	[0.515 , 0.689]	[-0.201 , 0.0757]	[-0.405 , -0.176]	[-0.316 , -0.083]
	PC1	PC2	PC3	PC4	PC5
Correlation	0.851	0.291	0.0674	-0.0716	0.279
95% CI	[0.761 , 0.937]	[0.14 , 0.441]	[-0.117 , 0.244]	[-0.209 , 0.0624]	[0.126 , 0.406]

Correlation coefficients between f_t and Fama-French five factors and the first five principal components of asset returns. We also report their 95% posterior credible intervals under the coefficient estimates. In Panels A, the cross-section of base assets used to estimate f_t includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios, while in panel B we employ the Kozak, Nagel, and Santosh (2020) cross-section of 74 characteristic-sorted portfolios.

IV.2 Implications for Structural Models

The reduced-form estimate of f_t 's risk price obtained in the previous subsection (Table 5) has direct implications for structural models, in particular, for the coefficient of relative risk aversion (RRA) and the implied risk-free rate. We consider two types of representative agent preferences: (1) additively separable CRRA and (2) Epstein-Zin (EZ) recursive utility – that separates RRA from the elasticity of intertemporal substitution (IES). In Internet Appendix O, we show how the estimated market price of consumption mean risk and our estimated dynamics for the consumption process can be mapped into utility parameters and preference-implied risk-free rate level and volatility.

Note that the estimated market price of consumption risk in Table 5 already generates a market risk premium in line with its historical average. Hence, the questions to ask are whether, given a preference setting, too high RRA is required to yield the estimated price of risk and whether the implied risk-free rate is realistic. That is, we ask whether our uncovered dynamics for the consumption process can jointly match the equity premium and risk-free rate puzzles.

Panels A–D of Table 7 report the results based on the aggregate consumption data. For CRRA preferences, we report the implied RRA level in the canonical C-CAPM ($S = 0$) and also using the ultimate consumption risk approach of Parker and Julliard (2005) ($S = 2, \dots, 14$), which allows consumption to react slowly to return innovations and back out

Table 7: Implications for structural models

	CRRA								Epstein-Zin	
	$S = 0$	2	4	6	8	10	12	14	$\psi = 1.5$	$\psi = 0.5$
Panel A. Aggregate consumption in 37 stock and nine bond portfolios (constant volatility)										
RRA	144.3	52.1	33.7	28.5	28.3	26.7	24.7	25.1	26.0	27.0
90% CI	[47.9, 234.9]	[17.3, 84.9]	[11.2, 54.9]	[9.5, 46.5]	[9.4, 46.1]	[8.9, 43.4]	[8.2, 40.3]	[8.4, 40.9]	[9.0, 41.9]	[10.1, 43.0]
$\mathbb{E}[r_f]$	29.40%	51.60%	39.03%	34.51%	34.32%	32.77%	30.87%	31.28%	1.71%	3.72%
$\sigma(r_f)$	103.27%	37.31%	24.11%	20.43%	20.29%	19.10%	17.69%	18.00%	0.48%	1.43%
Panel B. Aggregate consumption in 37 stock and nine bond portfolios (stochastic volatility)										
RRA	180.2	76.5	41.4	32.6	30.2	25.0	20.8	20.1	21.1	22.3
90% CI	[59.8, 293.4]	[25.4, 124.6]	[13.8, 67.4]	[10.8, 53.1]	[10.0, 49.2]	[8.3, 40.6]	[6.9, 33.9]	[6.7, 32.8]	[7.4, 34.0]	[8.6, 35.1]
$\mathbb{E}[r_f]$	12.86%	65.02%	46.60%	39.13%	36.88%	31.68%	27.28%	26.56%	2.08%	3.72%
$\sigma(r_f)$	192.96%	65.77%	32.62%	25.11%	23.10%	18.82%	15.51%	15.00%	0.49%	1.43%
Panel C. Aggregate consumption in Kozak, Nagel, and Santosh (2020) 74 anomaly portfolios (constant volatility)										
RRA	150.7	42.6	27.7	21.7	19.9	17.7	16.0	15.7	16.5	17.7
90% CI	[67.3, 235.4]	[19.0, 66.5]	[12.4, 43.2]	[9.7, 33.9]	[8.9, 31.1]	[7.9, 27.7]	[7.1, 24.9]	[7.0, 24.5]	[7.7, 25.5]	[8.9, 26.7]
$\mathbb{E}[r_f]$	44.65%	47.59%	34.42%	28.28%	26.30%	23.87%	21.83%	21.55%	1.77%	4.28%
$\sigma(r_f)$	126.13%	35.66%	23.14%	18.17%	16.65%	14.84%	13.35%	13.15%	0.56%	1.67%
Panel D. Aggregate consumption in Kozak, Nagel, and Santosh (2020) 74 anomaly portfolios (stochastic volatility)										
RRA	159.7	46.4	30.2	22.1	19.5	16.8	14.6	14.4	15.2	16.4
90% CI	[71.3, 249.4]	[20.7, 72.4]	[13.5, 47.2]	[9.9, 34.6]	[8.7, 30.5]	[7.5, 26.2]	[6.5, 22.8]	[6.4, 22.4]	[7.1, 23.4]	[8.3, 24.6]
$\mathbb{E}[r_f]$	41.64%	50.78%	37.09%	28.83%	25.96%	22.86%	20.29%	20.00%	2.16%	3.86%
$\sigma(r_f)$	158.52%	40.89%	26.18%	19.00%	16.71%	14.33%	12.43%	12.22%	0.61%	1.71%
Panel E. Shareholders' consumption in 37 stock and nine bond portfolios (constant volatility)										
RRA	110.1	17.9	11.0	10.0	9.8	7.0	7.4	5.7	6.5	7.7
90% CI	[36.5, 179.2]	[5.9, 29.1]	[3.6, 17.8]	[3.3, 16.2]	[3.3, 16.0]	[2.3, 11.5]	[2.4, 12.0]	[1.9, 9.4]	[2.6, 10.2]	[3.8, 11.4]
$\mathbb{E}[r_f]$	49.52%	23.58%	15.67%	14.46%	14.27%	10.81%	11.22%	9.15%	1.47%	5.51%
$\sigma(r_f)$	78.77%	12.78%	7.84%	7.13%	7.02%	5.04%	5.27%	4.11%	0.48%	1.43%

Estimates of RRA, average implied annualized risk-free rate, and its annualized volatility, using two cross-sections of asset returns: (1) 37 stock and nine bond portfolios and (2) Kozak, Nagel, and Santosh (2020) 74 anomaly portfolios. The risk price of f_t (and its 90% CIs), denoted by b_f , are from Table 5. Cumulative impulse responses of *nondurable* consumption growth, $\sum_{j=0}^S \rho_j$, are estimated using our MA model, with $\bar{S} = 14$ quarters. We consider specifications with and without SV. In models with SV, we assume three separate SV processes, $\sigma_{c,t}^2$, $\sigma_{f,t}^2$, and $\sigma_{r,t}^2$, all of which are allowed to drive conditional mean returns. In the CRRA case, the RRA coefficients are given by $\gamma_S = b_f / \sum_{j=0}^S \rho_j$, where $S \in \{0, 2, \dots, 14\}$ (see equation (IA.44) of the Internet Appendix). In the Epstein-Zin case, we consider two different values of IES, i.e., $\psi \in \{0.5, 1.5\}$, and $\gamma = (b_f + \frac{1}{\psi} \sum_{j=1}^S \kappa_z^j \rho_j) / (\rho_0 + \sum_{j=1}^S \kappa_z^j \rho_j)$ (see equation (IA.54) of the Internet Appendix). Mean and volatility of the risk-free rate are obtained using equations (IA.42) and (IA.48) of the Internet Appendix, where $\delta = 0.9965$ (following the quarterly calibration of Bansal, Kiku, and Yaron (2016, Table 6)). In Panels A–D, we consider the aggregate nondurable consumption data, while Panel E uses shareholders' consumption data.

the coefficient of risk aversion.²⁹ For EZ preferences, we considered two possible calibrated values for the IES, $\psi = 1.5$ and $\psi = 0.5$. Not surprisingly, when we consider only the contemporaneous response of consumption growth to asset return shocks ($S = 0$), the implied RRA estimate in the CRRA preference is larger than 100. As we allow for a slow reaction of consumption to asset return shocks, for example, $S = 14$, the implied RRA is much lower (from 25 in Panel A to 15 in Panel D), and values below 10 are now within the 90% confidence bands. However, the implied real risk-free rate is too high (20–31%) and volatile (12–18%) with CRRA preferences.

Does our method improve the performance of the CRRA model compared to Parker

²⁹See equations (IA.43)–(IA.44) of the Internet Appendix.

(2001)? Figure IA.36 of the Internet Appendix plots the RRA coefficients implied by the VAR approach of Parker (2001). Similar to our state-space formulation, the VAR-implied RRA declines from more than 100 to 28 at the three-year horizon. However, the RRA coefficients based on our method can be significantly lower, particularly in the cross-section of KNS anomalies. What drives the difference between VAR and our MA model? The former assumes that shocks to the market index are the *only* spanned innovations in consumption. Instead, our MA model, as we show in simulations, and which seems to be the case in the data (see Table 6), can capture an arbitrary linear combination of systematic latent factors spanned by financial markets. Hence, when consumption growth reacts to other factors beyond only the market return (as in the cross-section of KNS anomalies), we detect a significant difference between the VAR- and MA-implied estimates of RRA.

EZ preferences in Panels A–D of Table 7 yield moderate levels of risk aversion (15 – 27), with a substantial likelihood of values below 10, irrespective of the calibrated IES coefficient. Instead, the risk-free rate level and volatility are quite sensitive to the choice of ψ . For $\psi = 0.5$, the implied level (3.7 – 4.3%) overshoots the historical average (1.68%), and the implied volatility (1.43 – 1.71%) roughly matches its historical, *ex-post*, volatility (2.44%).³⁰ For $\psi = 1.5$, EZ preferences almost perfectly match the risk-free rate level (1.71 – 2.16%) and provide reasonable values for its implied, *ex-ante*, volatility (0.48 – 0.61%).

Finally, in Panel E of Table 7, we compute the implied RRA coefficient and risk-free rate based on the impulse responses of shareholders' consumption growth (in Figure 4). In this case, due to the much larger responses of shareholders' consumption to return shocks, the RRA coefficients are about 6 – 8 for $S = 14$ and EZ preference.³¹ Nevertheless, only EZ preferences with $\psi = 1.5$ can also match well the observed risk-free rate.

V Conclusions

We identify the stochastic process of consumption growth using the information contained in financial returns. Our strategy relies on the central insight of the intertemporal Euler equation of models that have consumption as one of the state variables entering the utility

³⁰Note that this historical volatility is an upper bound for the *ex-ante* real risk-free rate volatility that the models should match.

³¹A caveat is that the risk price estimate, b_f , is based on a larger sample (1963:Q3–2019:Q4), whereas the IRFs of shareholders' consumption growth are based on a limited sample (1982:Q1–2004:Q3).

function: Most shocks affecting the household force it to adjust both investment and consumption plans. We confirm this mechanism by using the micro-data on shareholders' and non-shareholders' consumption from Malloy, Moskowitz, and Vissing-Jorgensen (2009).

We show that a flexible parametric model with common factors driving returns and consumption identifies the slow-varying conditional mean of consumption growth. This component is persistent at business cycle frequency and is economically substantial, capturing more than a quarter of the variance of consumption growth. Furthermore, the predictability that we uncover does not appear to be caused by measurement issues of consumption such as time averaging, measurement error, and benchmarking. Hence, our findings indicate that the shocks spanned by financial markets are first-order drivers of consumption risk.

The conditional consumption mean shocks carry a significant risk premium in the cross-section of asset returns, and demand an annualized Sharpe ratio of about 0.5. Furthermore, we embed the estimated consumption dynamics into an otherwise standard Epstein and Zin (1989) preference setting with elasticity of intertemporal substitution above one (as in Bansal and Yaron (2004)). Such a calibration can match both the unconditional equity premium and the risk-free rate level and volatility, with low relative risk aversion.

We also show that detecting time variation in consumption volatility requires properly accounting for both the conditional mean process that we uncover and the measurement issues in consumption data. Nevertheless, we find no empirical evidence for the stochastic volatility of consumption driving time-varying risk premia.

Compared to the previous literature, our findings are obtained using not only more general and flexible specifications but also less restrictive priors and much richer asset return data. In addition, we provide a sizable set of complementary evidence that does not rely on our parametric formulation but rather supports the main findings of our state-space analysis.

Our findings have first-order implications not only for macro-finance models but also, and arguably more importantly, for the assessment of the costs of business cycle fluctuations and optimal fiscal and monetary policies. We defer the study of these effects to future work.

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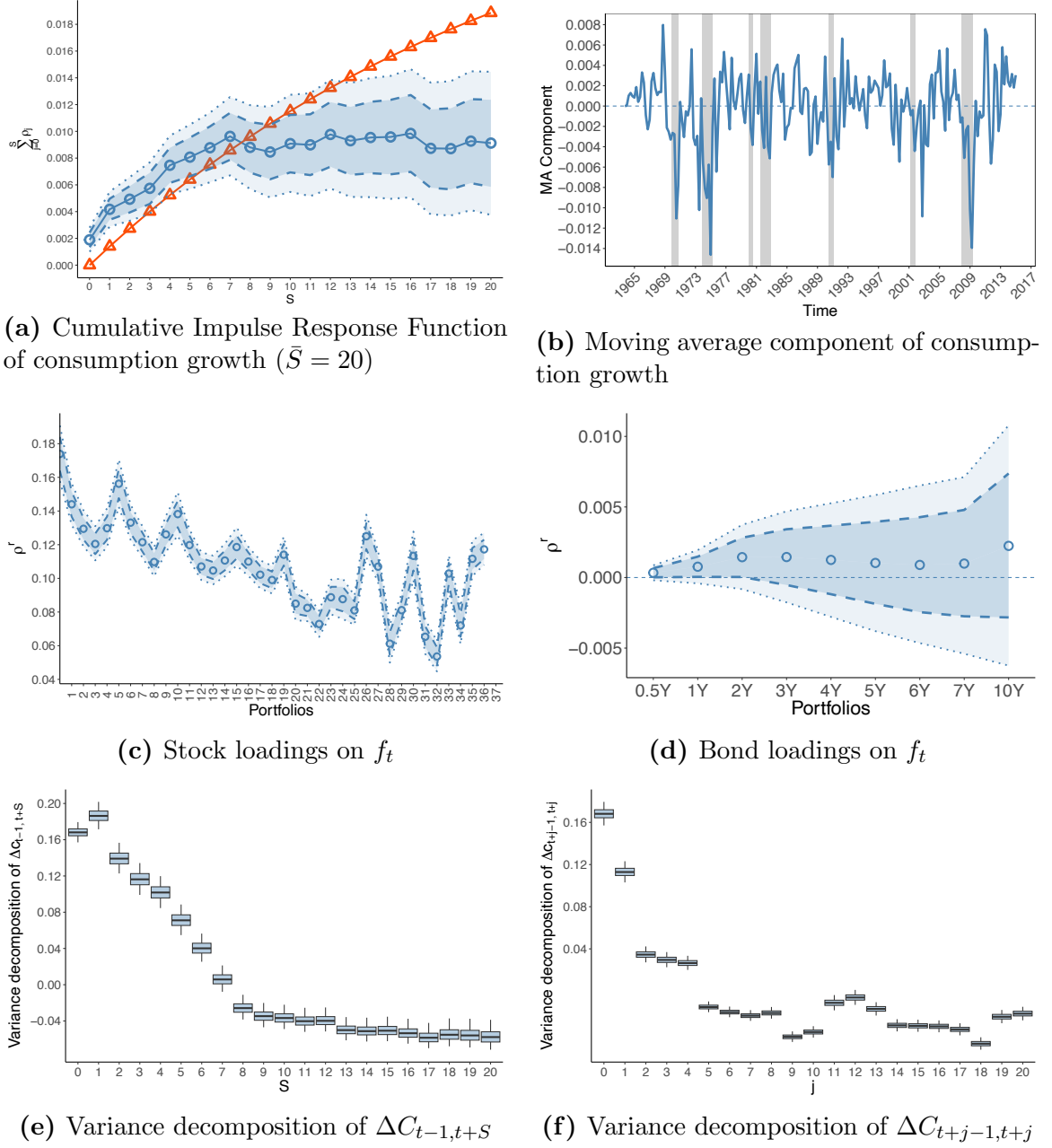
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Figure A1: Robustness check: $\bar{S} = 20$.



The figure plots (a) the cumulative IRF of consumption growth ($\bar{S} = 20$), (b) the MA component of consumption growth, (c) stock loadings on f_t , (d) bond loadings on f_t , (e) variance decomposition of $\Delta C_{t-1,t+S}$ (adjusted R^2), and (f) variance decomposition of $\Delta C_{t+j-1,t+j}$ (adjusted R^2). We consider a single-factor model in equations (2)–(3), with $\bar{S} = 20$. The cross-section of test assets includes 25 size-and-value-sorted portfolios, 12 industry portfolios, and nine bond portfolios. Red triangles in Panel (a) denote the cumulated IRF of the long-run risk model (Bansal and Yaron (2004)) in the monthly calibration of Hansen, Heaton, Lee, and Roussanov (2007) aggregated to quarterly frequency.